

User Manual for IOS Integrated Optimisation Suite by



Document history

Version:	Published:	Author:	Remarks:
V0.1	02/08/2011	Guy op de Beeck	
V0.3	08/08/2011	Guy op de Beeck	
V1.0	09/08/2011	Felix Franz	
V1.1	06/09/2012	Guy op de Beeck	
V1.2	26/08/2013	Guy op de Beeck	Draft

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1. Solution Description

The solution description consists of a High Level architecture diagram and the interface descriptions. As this is specific for every implementation, this information has been separated into a different document:

Please view separate document: CellSens_Architecture_<Operator>.doc

2. Functional Description

2.1 Start Up Phase



Figure 1 Startup splash screen

At the start up (Figure 1) of the IOS application, the following steps are taken before the program is accessible:

2.1.1 Version check

IOS has a built-in auto-update feature. At startup the first action of the program is to check on the AutoUpdate Server if a new version is available. The AutoUpdate server is defined in the IOS.exe.config file in the root of the IOS program folder under the key "AutoUpdate Server". When a new version is found, the existing application will close and a (HTTP) download session will start, overwriting or adding files to the existing version.

Once download is finished, the new version of IOS is ready to be started.

Error State:

If the AutoUpdate Server is unreachable, a warning will be given but the program will continue to boot up. It is recommended though to ensure a connection to the internet and to verify with support@cellsens.com if AutoUpdate Server address is correctly implemented.

Options:

- Enable/Disable: The Version Check and AutoUpdate feature can be disabled manually. The IOS.exe.Config contains the key "VersionCheck". The default value for this key is "1". If set to "0" the version check and auto update feature will be disabled.
- Proxy Server: if a proxy server is required to access any HTTP server, then the value of this key should be set to "1". Default value is "0"

2.1.2 License Check

A license is verified based on 2 parameters: the Windows Username and the current date. The license check process will verify if the particular Windows Username has a valid license.

Depending on the implementation, a license check is either "internal", meaning it doesn't require external internet connection, or "external", meaning it will check license using an internet connection.

- Internal

If the system is configured for an internal check, the client will verify validity of the license on the IOS Server.

- External

A secure SSH connection is made to the external IOS License Server using port 22 and port 3306. The IOS License Server IP address is defined in the IOS.Config.exe file in the root of the IOS program folder under the key "CellSens Server".

During the license verification it is checked if the Windows UserID is registered on the license server for the company to which the license has been granted.

Error State:

The failure of the license check can be due to following reasons:

- no connection can be made to the server
- the license date expired
- the Windows UserID supplied is not registered in the license server

A failure if this steps results in the shutdown of the IOS application. It is recommended to contact support@cellsens.com.

2.1.3 IOS Server Connection

The IOS Server is the heart of the IOS application. The Smart Client verifies if the connection to the IOS Server is active. The standard requirement is a valid ODBC connection based on MS SQL named "IOS_Server". The verification consists of opening the data connection towards IOS_Server database on the MS SQL server defined in the ODBC DSN.

Error State:

If connection to the IOS_Server fails, the program closes automatically. Ensure that the ODBC connection and naming is correct.

2.1.4 IOS Configuration

The IOS Server contains the configuration information for the smart client. During this step, the configuration is downloaded and applied to the smart client.

2.1.5 Datasource Verification

The IOS program makes use of different datasources in the network as described in the Solution Architecture of this document. Each of the datasources is verified if they are operational. This is done based on opening each ODBC interface defined in the Solution Architecture (Table 1.).

Error State:

An error message pops up if:

- no connection can be made to the server

Although the program continues booting up, some functionality might not be operational anymore. Check the correct naming of the ODBC interfaces and contact support@cellsens.com.

2.1.6 Progress Bar

During the progress bar phase, the IOS Smart Client downloads object tree information from the OSS.

Error State:

Usually if the Data Verification phase failed, it is likely that the same error message is popping:

- no connection can be made to the server

Although the program continues booting up, some functionality might not be operational anymore. Check the correct naming of the ODBC interfaces and contact support@cellsens.com.

The program booted successfully if two windows forms are displayed on the main screen: The Map windows form and the Console windows form.

Options:

- Enable/Disable ObjectTree reload

The key “ReloadOnStartup” in the IOS.exe.config toggles if at startup a new set of object tree information needs to be loaded. By default this is set to “1”. If a user wishes faster boot times, it can be set to “0” to make use of the existing locally stored objecttree information. If set to “0”, each object tree will contain information on how old the tree is. Right clicking the object tree and selecting the “Reload” option, will trigger a reload of the objecttree

2.2 IOS – Map Window

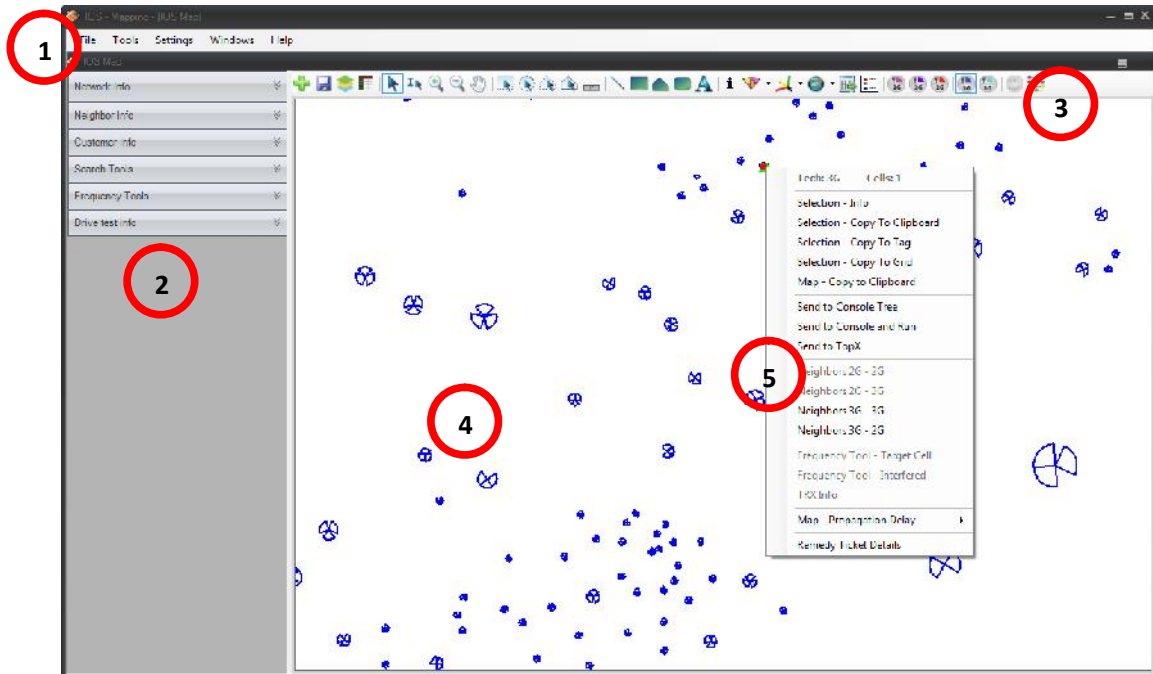


Figure 2 The IOS Map Window form

The IOS Smart Client is based on a form specialized in GIS related functions called the “IOS Map” and a form specialized in charting and performance management functions called the “IOS Console”.

The Map Window functionality can be split in five parts:

1. Menubar
2. Explorerbar
3. MapToolbar
4. Mapping frame
5. Mapping context menu

2.2.1 IOS – Map – Menubar

2.2.1.1 File Drop Menu

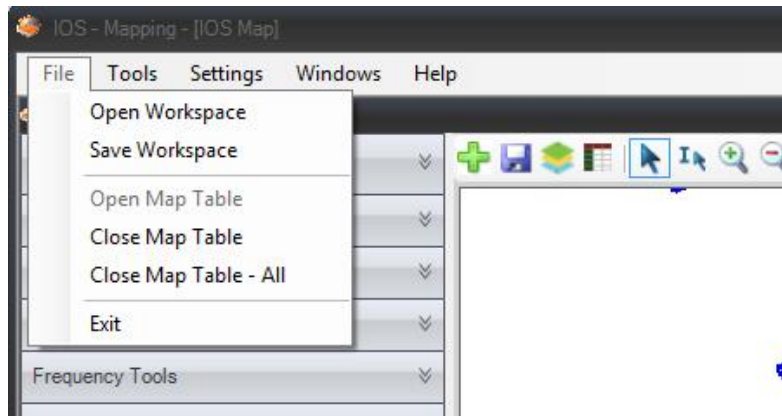


Figure 3 File Drop Menu

Open Workspace

Open Workspace functions pops up a File Dialog box which allows you to open IOS Workspace files with extension .MWS. The action will clear the existing mapping workspace, open the tables defined in the workspace file and set the view according to the settings in the MWS file.

Save Workspace

Saving the workspace pops up a Save Dialog box in which you have to define the filename of the workspace. The default location for storing the file is: "C:\Program files\IOS\Data".

The workspace file is an XML document (.MWS extension) that contains the locations, descriptions and metadata of all the maps, tables, layers and settings that make up the current mapping workspace.

Close Map Table

The function pops up a window showing all the currently open files in the mapping environment. To close a table, select a record in the window and confirm the close.

Close Map Table - All

All tables residing in the mapping environment will be closed.

Exit

Exits the application, closing all windows.

2.2.1.2 Tools Drop Menu

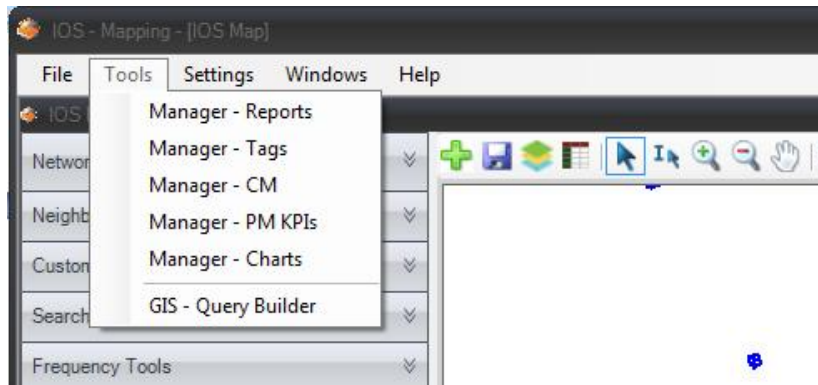


Figure 4 Menubar - Reporting

Manager - Reports

Launches the Reports Manager. This module allows for creation of PowerPoint slides based on the charts gathered in the Console. More information can be found in chapter 2.2.5.

Manager – Tags

Launches the Tags Manager. This module enables the user to create groups of cells and label them with a Tag. This Tag can then be used as a reference in the statistics Console. More information can be found in chapter 2.2.6.

Manager – CM

Launches the Manager for Configuration Management data. For more information I refer to chapter 2.2.7.

Manager – PM KPIs

Launches a form to manage KPI's related to the Performance Management. For more information I refer to chapter 2.2.8.

Manager – Charts

Launches a form to manage all the charts related to the Performance Management. For more information I refer to chapter 2.2.9.

GIS – Query Builder

Launches a form that enables the user to maximize the use of the powerful Mapxtreme/Mapinfo compatible spatial query engine. For more information I refer to chapter 2.2.10.

2.2.1.3 Settings Drop Menu

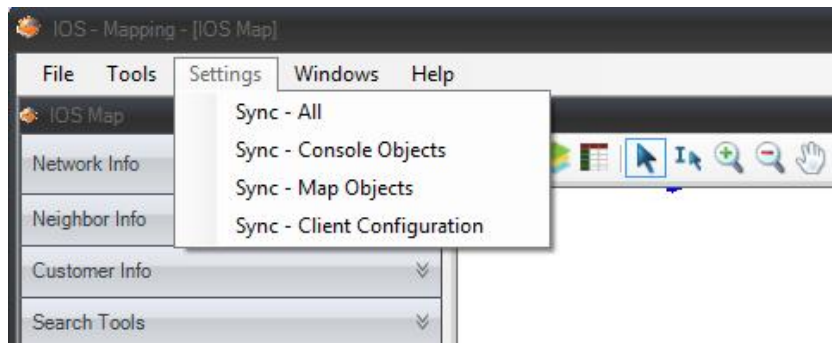


Figure 5 Settings Menu

Sync – Console Objects

The objects in the object tree are either loaded on start-up of the application and is not in live connection with data source. The Sync-Console objects method enables to update all the object trees to the latest data.

Remark: it is also possible to update an individual object tree by right clicking on the object tree and selecting "Reload".

Sync – Map Objects

This is Obsolete

Sync – Client Configuration

Most configurations of IOS are stored on the IOS Server. This function is to refresh the configuration information from the IOS Server towards the client. It is used to avoid a reboot when a new configuration is applied.

2.2.1.4 Windows Drop Menu

This is Obsolete

2.2.1.5 Help Drop Menu

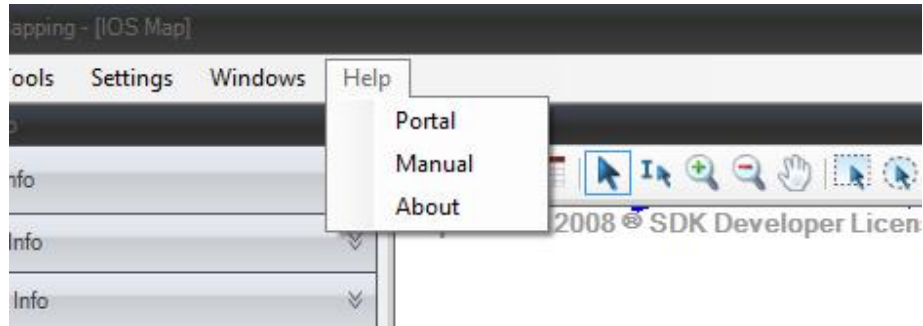


Figure 6 Help Drop Menu

Portal

The portal menu item is a hyperlink to the CellSens Web Portal. The Web portal is a web interface allowing for reporting back problems and improvement proposals.

For login and password, contact your administrator.

Manual

This menu item is a PDF hyperlink to this document

About

Pops up a Dialog Box showing the version number and license information.

2.2.2 IOS – Map – ExplorerBar

2.2.2.1 Network Info



Figure 7 Network Info Explorer Bar Item

The network info consists of 4 elements: a date selection field, a “Get Networks” button, a “Network State” button and the status bar.

The date selection field indicates for which dates the network configuration files are present. Each night, the IOS Server queries the planning database (see architecture document) and stores the relevant information. To fetch the network data, select a bold date and press the “Get Networks” button.

The status bar will show the progress of the network generation. Each technology layer (EGSM, DCS, 3G F1, 3G F2, ...) has first a download session and then a geometry creation session. The result is saved as .TAB files in the folder:

“C:\Users\<windows user name>\AppData\Roaming\CellSens\IOS\<deploymentname>”

The naming scheme for network configuration layer is “Cells_<TECHNOLOGY>” with <TECHNOLOGY> being eg. EGSM, DCS, WCDMA_F1, WCDMA_F2, etc..

After the generation, the layers are automatically added to mapping environment and made active by toggling the “Quick Layer” buttons (See chapter 2.2.3).

The (optional) “Network State” button provides network status layers on top of the network configuration layers. The data is fetched from the live OSS for each technology and saved internally following the naming “<network config layer name>_Status” and “<network config layer name>_Tickets” (for example: Cells_EGSM_Status). These layers are automatically mapped with a specific thematic based on the “status” field

- Green Fill: Site is ok (status = 1, shown on figure 6 next to circle 1)
- Orange Fill: Site has alarms (status = 2, shown on figure 6 next to circle 2)
- Red Fill: Site is locked (status = 3, shown on figure 6 next to circle 2)
- Orange Outline with stripes: Site has an open Ticket in database. (Circle 4)
- No fill: indicates that site has no status information (Circle 5)

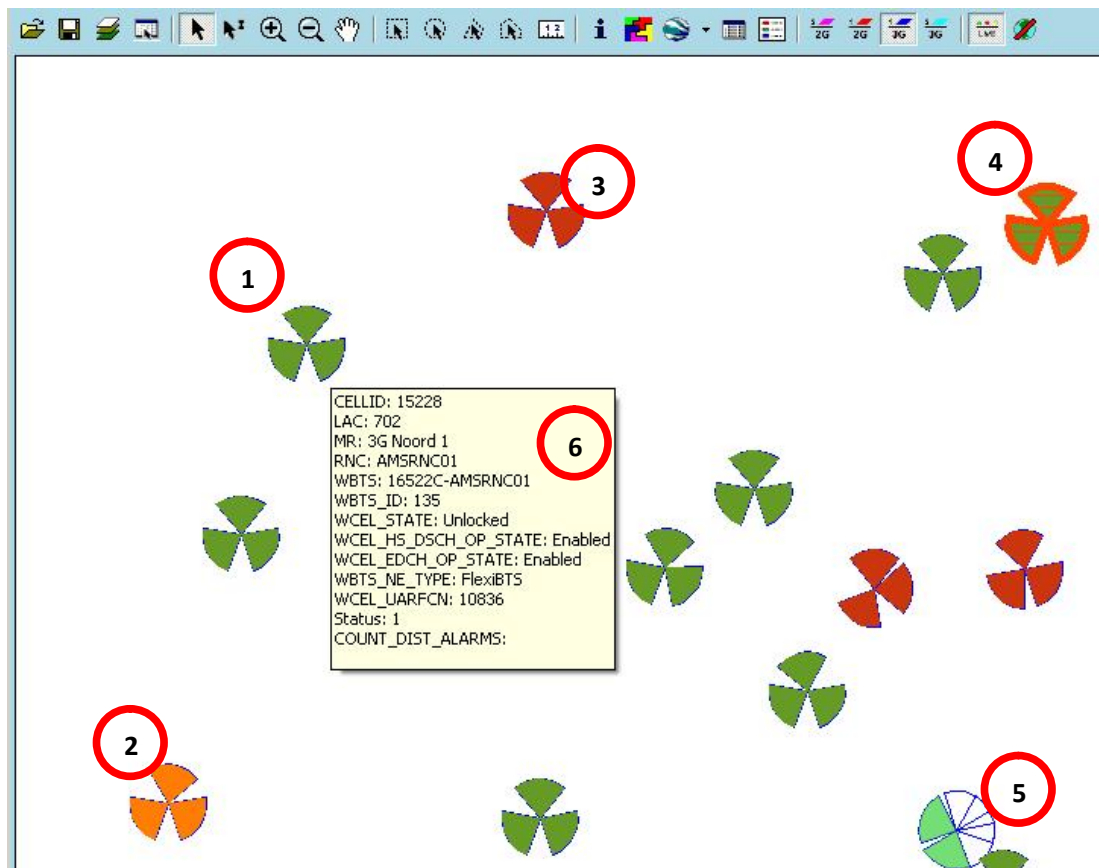


Figure 8 Network State

The data is a snapshot of the current situation and doesn't change dynamically. If new status information is required, the “Network State” button has to be pressed again.

Tooltip Fields of Network State (Circle 6)

Field	States	Description
WCEL_State	Locked/Unlocked	
WCEL_HS_DSCH_OP_STATE	Enabled/Disabled	HSPA Downlink Channel active
WCEL_EDCH_OP_STATE	Enabled/Disabled	HSPA Uplink Channel active
WBTS_NE_TYPE		Type of NodeB
WCEL_UARFCN		The radio channel of the WCEL
Status	1,2,3	1. Status OK 2. HSPA inactive 3. Locked

Table 1 Network State fields description

To hide and show the Network State Thematic Layers and Tickets Layers quickly, the “Network Status” quick button on the mapping toolbar can be used. More information is available about this button in chapter 2.2.3.

2.2.2.2 Neighbor Info

The image shows the 'Neighbor Info' Explorer Bar Item interface. It is a vertical panel with a yellow header. The interface is organized into two main sections: 'Basic Settings' and 'Optional Settings', indicated by brackets on the right. The 'Basic Settings' section includes:

- Relationships:** Radio buttons for 2G - 2G, 3G - 3G (selected), 2G - 3G, and 3G - 2G.
- Direction:** Radio buttons for Incoming and Outgoing (selected).
- Layer:** Radio buttons for 3G Intrafreq (selected) and 3G Interfreq.
- Definition:** Radio buttons for Defined (selected) and Undefined.
- Statistics:** Radio buttons for Stats On and Stats Off (selected).

 The 'Optional Settings' section includes:

- Cell ID:** A text input field.
- Days:** A numeric input field with a value of 1.
- Get:** A button to retrieve data.
- Clear:** A button to clear the data.
- Arrows:** A button to toggle arrow display.
- Parameter:** A dropdown menu currently showing 'NB_ATT'.
- TopX:** Two sliders for values 10 and 15.
- Sort Desc:** A checkbox that is checked.
- Cell:** A slider for cell selection.

Figure 9 Neighbor Info Explorer Bar Item

Expanding the Neighbor Info Explorer Bar item shows the available options when querying for neighbor definitions and statistics.

There are two methods for displaying the neighbor information.

Workflow – Neighbor information through Explorer Bar:

- Set the technology
- Set the direction
- Set the inter or intra frequency in case of 3G technology
- Set defined or undefined (undefined is only available in 3G).
- Set Statistics ON or OFF. To ensure high speed querying, a distinction is made between the queries towards the database. If Stats OFF is selected, only the definitions and types will be fetched. If Stats ON is selected, the following statistics are generated:
 - o 2G: Neighbor Attempts, Neighbor Success, Neighbor Failures, Percentage Neighbor Failures
 - o 3G: Neighbor Attempts, Neighbor Success, Neighbor Failures, Percentage Neighbor Failures, EcNo Difference, Ecno, RSCP

Fetching statistics can be slow. Default value is OFF.

- Enter the Cell ID for which the statistics are wanted
- Enter the number of days (default is 1).
- Press Get:
 - 1** o Query is fired and result is mapped as arrows in the mapping window. Hovering over an arrow with the mouse shows the details of a neighbor through a tooltip.
 - 2** o The label of the arrow is the number of handover attempts.
 - 3** o If statistics is ON, then a graduated thematic is applied on the Attempts, showing a thick red line for high attempts, and thin black line for low attempts. The legend can be called on the maptoolbar (see 2.2.3)
 - o Yellow fill indicates a target cell whereas a Green fill indicates the source cell. Orange indicates Undefined NB, only applicable in 3G.
 - 4** o All information is also stored in a grid that can be called from the maptoolbar button “Show Grid”.

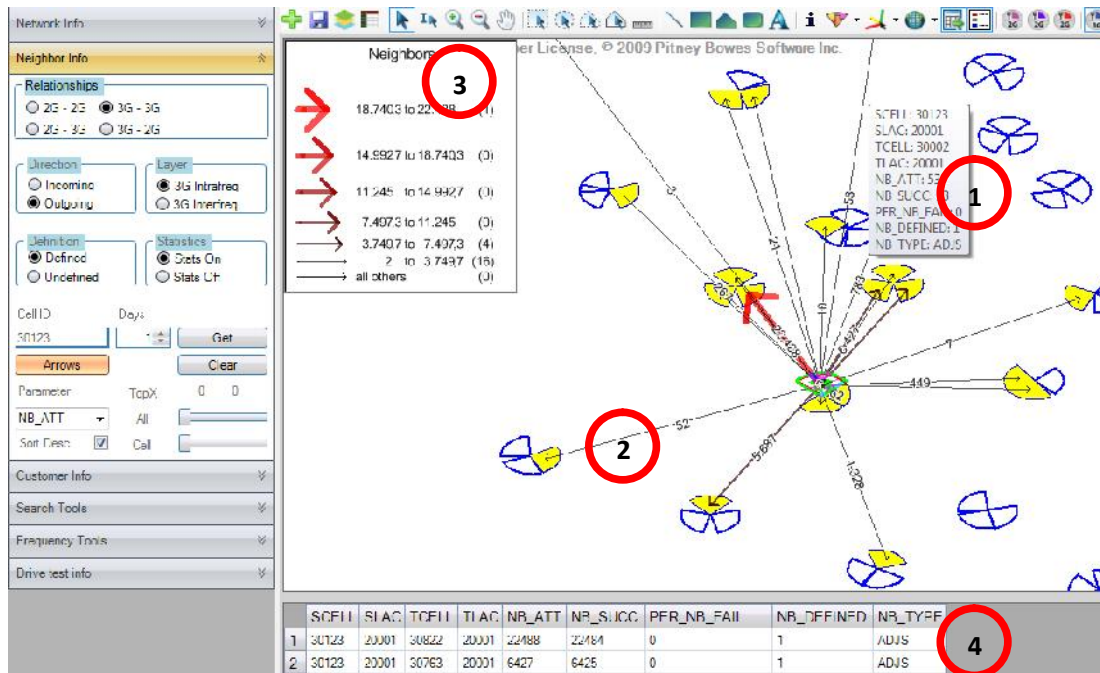


Figure 10 Neighbor Mapping

Workflow – Neighbor information through Map Right Click

- Select the cell of interest on the map
- Right click and choose the neighbor relation type
- The query will be fired and the result will be mapped based on the settings in the explorer bar.

The query result is stored as tables in the mapping environment with following naming:

- NB_<relation>_Source (Example: NB_3G3G_Source)
- NB_<relation>_Target

After mapping has been conducted, changes applied to the direction settings, inter and intra frequency settings and defined/undefined settings will be immediately reflected without requiring a new query action on the server.

If Stats ON/OFF selection is changed or other settings are changed except mentioned above , after the mapping has been finished, one of the workflows need to be repeated.

The clear button removes all the Neighbor Layers in the mapping environment.

Description of the neighbor fields:

Field	Relation	Description
SCELL	All	Source Cell ID
SLAC	All	Source LAC
TCELL	All	Target Cell ID
TLAC	All	Target LAC
NB_ATT	All	Attempts from source to target
NB_SUCC	All	Successes from source to target
NB_FAIL	All	Failures from source to target
PER_NB_FAIL	All	Percentage failures
NB_ECNO_DIFF	3G/3G	
NB_ECNO	3G/3G	
NB_RSCP	3G/3G	
NB_DEFINED	All	1 if Neighbor is defined, 0 If neighbor is undefined
NB_RSSI	3G/2G	In 3G2G
NB_BSIC_VERIF_TIME	3G/2G	
NB_TYPE	All	ADCE/ADJS/ADJI/ADJW/ADJG

Table 2 Fields Description of NeighborResultset

Layer Control

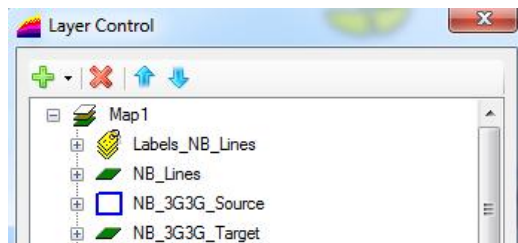


Figure 11 Neighbor Info - Mapinfo tables

Each query adds default named tables to the mapping environment:

- NB_Lines: represents the Mapinfo table holding the lines and data of the relationship
- NB_3G3G_Source: represents the Mapinfo table holding source cell data
- NB_3G3G_Target: represents the Mapinfo table holding the Target cell data
- Labels_NB_Lines: the configuration to visualize the label on the lines.

Each of the Mapinfo tables can be called into the Grid using the appropriate maptoolbarbutton as described in 2.2.3.

Optional Settings

The Optional Settings in the Neighbor Explorer Bar allow for more advanced filtering on the NB query.

Process:

- Select the parameter for which you want to apply the filter. By Default this is set to ATT for Attempts.
- Check or Uncheck the “Desc checkbox”. This enforces a sort before the filtering. When the user wants to identify the top 5 neighbors based on attempts, the sort order needs to be set to DESC.
- The ALL sidebar setting represents the number of records that needs to be returned in the entire query. If set to 0, it means that all records will be returned and no filtering is applied.
- The Cell sidebar setting represents the number of neighbor relations of each cell that needs to be drawn. This is only valid for a multicell-neighbor query. If set to 0, no filtering is applied.

Example 1: Visualize Top 10 NB relations from 1 cell:

- Set Stats ON
- Set Parameter to NB_ATT
- Check Sort Desc
- Set All Slidebar to 10
- Select Cell of interest on map
- Press “Get” button

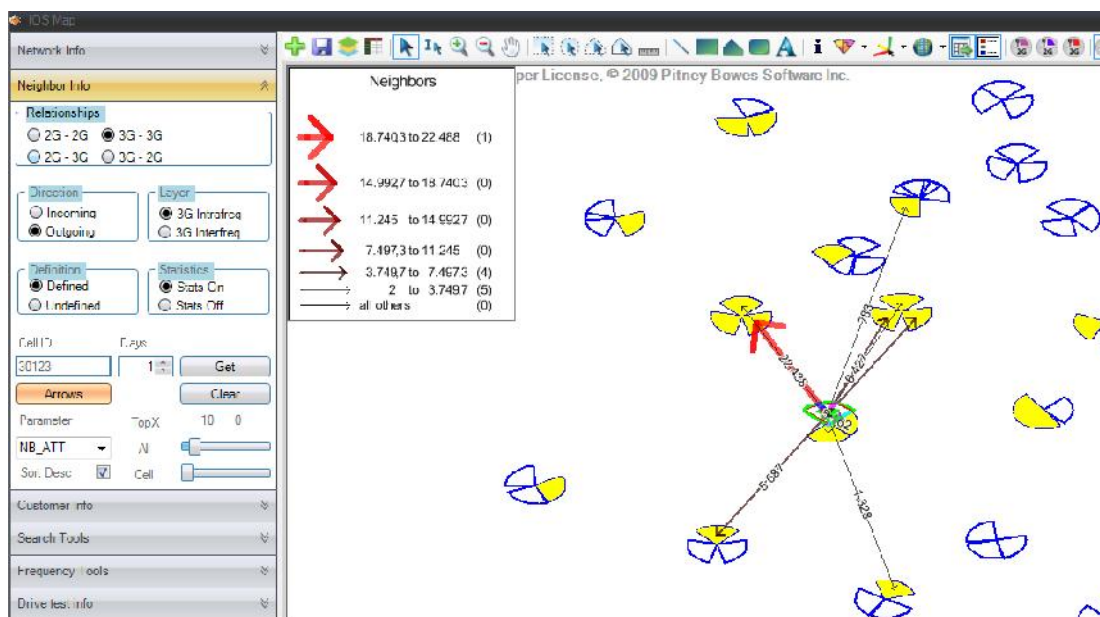


Figure 12 Neighbor Info - Filter Sample

Example 1: Visualize Top 10 NB relations from a 2 cell selection:

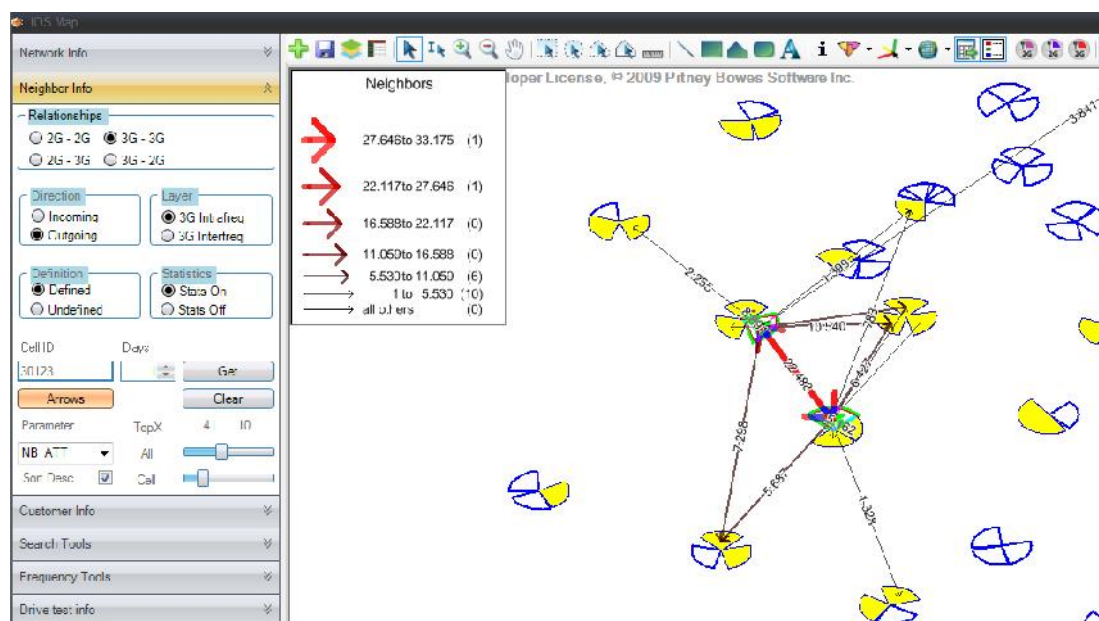


Figure 13 Neighbor Info - MultiCell Sample

This selection takes from the 41 most attempted NB relations, the top 10 of each selected source cell.

2.2.2.3 Customer Info

The screenshot shows the 'Customer Info' window in the IOS Map application. The window contains several input fields and a 'Get' button. The fields are: IMSI, MSISDN, Call Type (with radio buttons for CS Information and PS Information), Start Time (13/08/2013 15:41), and End Time (27/08/2013 15:41). The 'Get' button is at the bottom right.

Figure 14 Customer Info Explorer Bar Item

The Customer Info feature enables the user to visualize the activity and events of a specific customer in CS and PS domain. The data source used is a Call Data Record (CDR) database.

Workflow – Get Customer Activity Info

- Enter the IMSI of the customer
- Set if you require CS or PS info
- Set the start and end date for the monitoring
- Hit “Get”

A query will be fired towards the CDR database. The result will be stored in the grid, accessible through the maptoolbar “Show Grid” button, and will be mapped with two types of symbols:

- Cyan Gradient Circles: the location of the circle indicates where a user has been active. The size of the circle indicates the frequency of the cell where the user has been active. A large circle hence shows where the customer is active most of the time.
- Red Stars: Indicates the cell in which a user experienced a problem

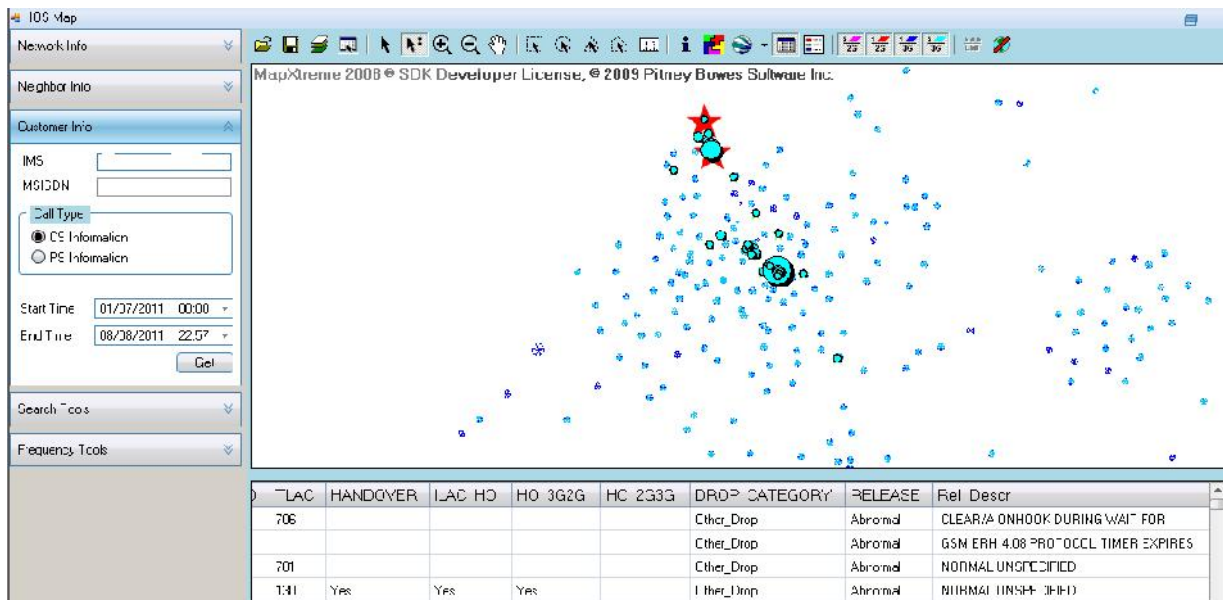


Figure 15 Customer Info Result

The following tables are generated:

- o Customer_Grouped_Map_Abnormal
- o Customer_Grouped_Map_Normal

Tip:

Double Clicking on the row of the map grid, will center the map on the “ocell” identifier of that row.

2.2.2.4 Search Tool

The screenshot shows the 'Search Tools' window with four sections, each with a red circle and a number next to it:

- 1** Points to the 'GeoCoder' section, which has a 'Geo Location' text input field and a 'Search' button.
- 2** Points to the 'Object' section, which has 'Cell ID' and 'Site ID' text input fields, a 'LAC' dropdown menu, and 'Search' buttons for each.
- 3** Points to the '2G' section, which has a 'Channel' text input field, an 'Auto Zoom' checkbox, and radio buttons for 'All', 'BCCH', 'TCH', and 'BCCH + BSIC' (which has an associated text input field).
- 4** Points to the '3G' section, which has a 'UARFCN' dropdown menu (showing '10836') and a 'Search SC' text input field with a 'Search' button.

Figure 16 Search Tool Explorer bar item

- 1** The Search explorer bar item contains of a geocoder and special searching functions specifically designed for cell information.

The geocoder is a flexible geographic information search engine. It can find the location of addresses, postal codes or clearly described points of interests. The search only returns one results, displayed as a red cross on the map. The map is automatically centered on the red cross.

- 2** Object search group box allows to search Cell objects and Site objects that are currently actively displayed (layer quick buttons must be toggled in the maptoolbar). Searching a cell is done through entering the cellid and clicking search. If the cellid has multiple LACs, the option is presented to select the LAC of interest. The result is highlighted and the map is centered on the selected object.

Searching a site is done by entering a site id. The result will highlight the three cells of that site.

The source is based on the Cells_<Tech> tables built from “Get Networks” action.

3

The third groupbox are channel searching functions specifically for 2G. Entering a valid channel number and pressing search button will fire a query to the relevant database (OSS or CM data warehouse). The results will be displayed on the map (See Figure 10):

- Green dots: Channel number matches with search
- Yellow dots: Channel number is adjacent to channel searched

The result is stored in the table: ARFCN_Search.

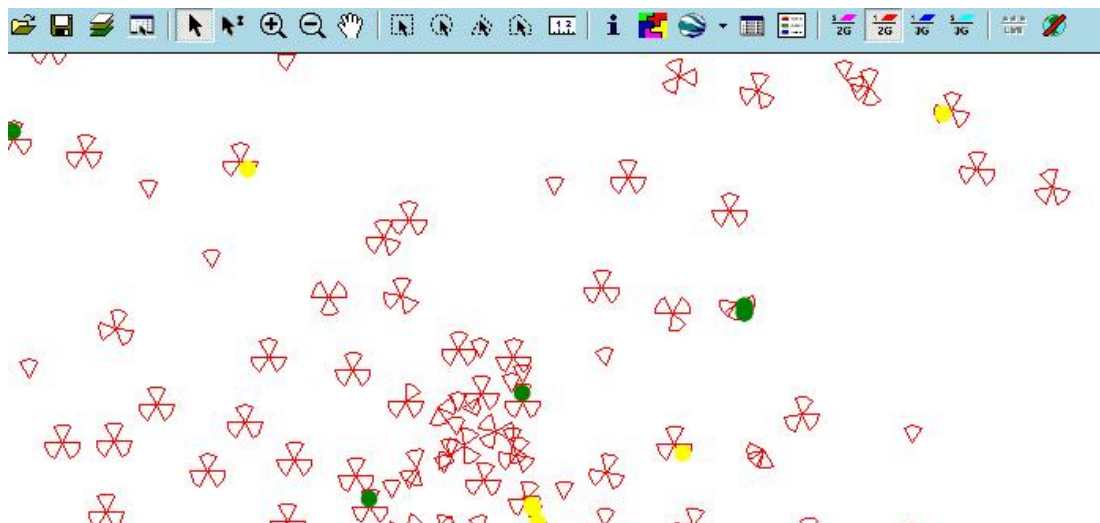


Figure 17 2G Channel Search output

The groupbox enables additional filtering:

- All: No filter is applied, all channels used are taken into account
- BCCH: search only conducted on BCCH channels
- TCH: search is only conducted on TCH channels
- BCCH+BSIC: search is only conducted for BCCH channel and BSIC. The radiobutton activates when a value is entered into the BSIC textbox.

4

Scrambling code searching in 3G is conducted by entering the scrambling code, selecting the UARFCN of interest and pressing search. A query will be fired towards Live OSS and the result will be mapped by highlighting the sectors where the scrambling code is detected.

The table in the mapping environment is called: “SC_Search_Map”.

2.2.2.5 Frequency Tool

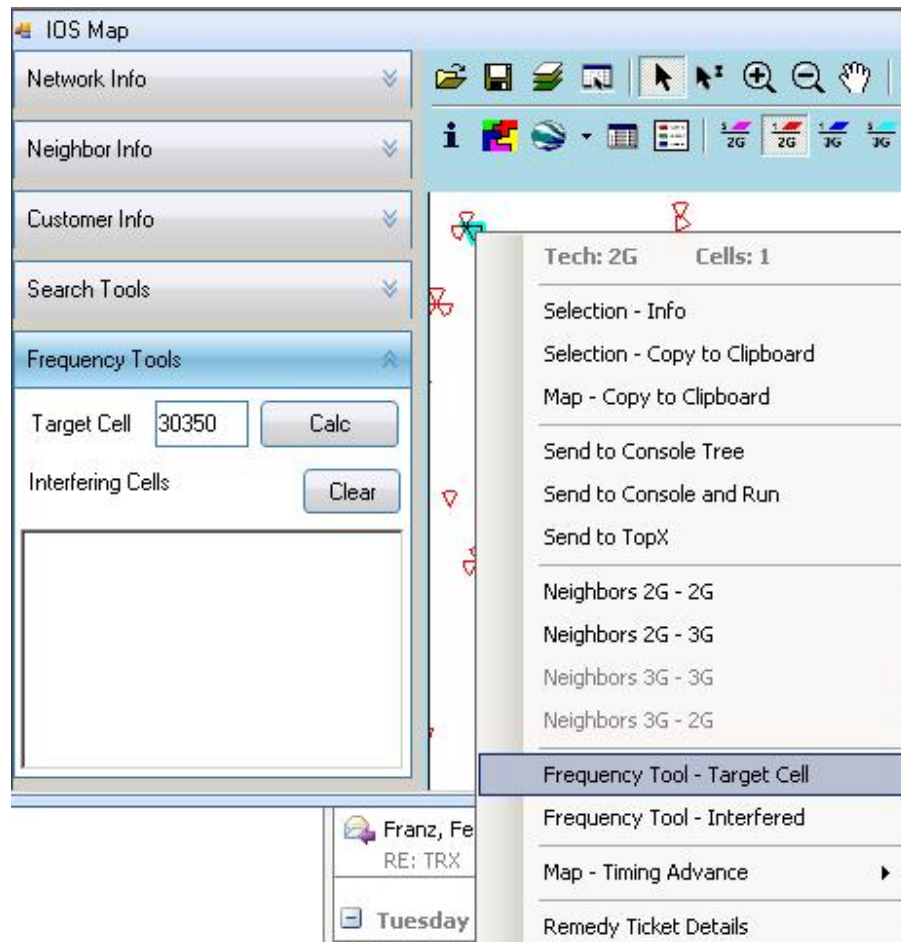


Figure 18 Frequency Tool Explorer Bar Item

The Frequency Tool is a 2G specific tool to facilitate the identification of a free channel based on a selection of cells and the defined Neighboring cells.

Workflow – Find a free frequency

- Identify the target cell for the free frequency
- Right click on that cell and choose “Frequency Tool – Target Cell”. The CellID will filled into the text box named Target Cell.
- Select a group of cells around the target cell and right click to choose the “Frequency Tool – Interfered” option. It is allowed to have the Target Cell in the Interfered selection.
- Now press Calc to produce the results table as shown in Figure 12.
- Find a ARFCN with 0 Occurrences that is not a Guard Band.

- Double click on that row to visualize the nearest co-channel and adjacent channels

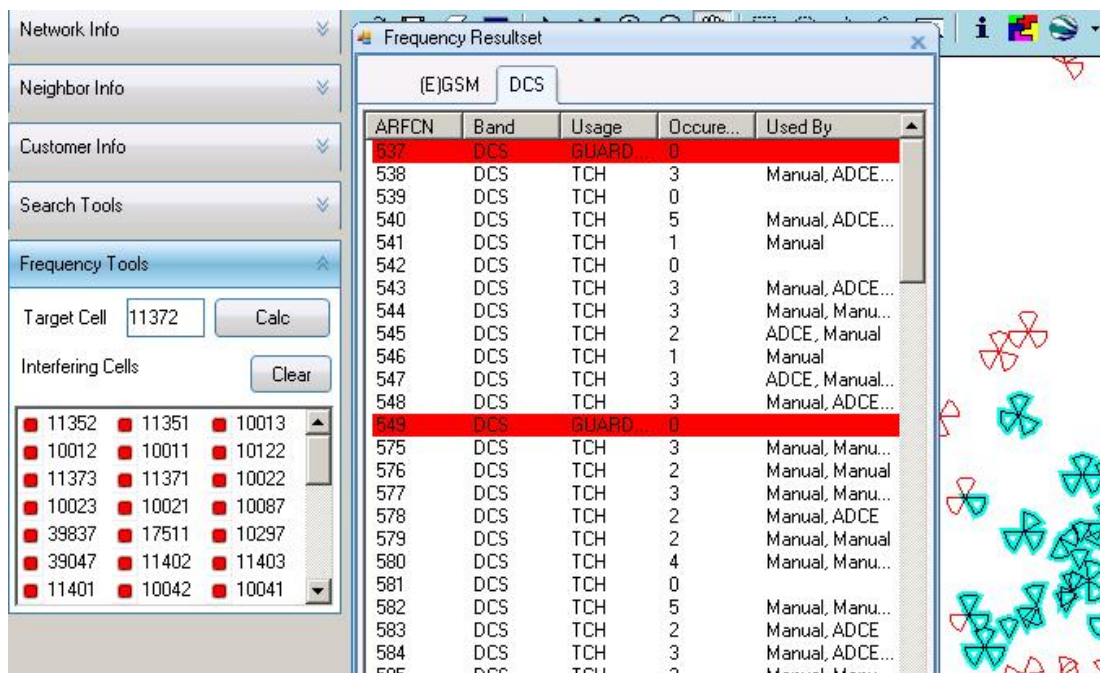


Figure 19 Frequency Search result

Two tabs are available in the resultset: (E)GSM and DCS. Per band, an overview has been created on how frequently channels are used in the selection made, but also taking into account the frequencies of the defined neighbors of the target cell.

Field	Description
ARFCN	Radio frequency channel number
Band	The band of the ARFCN ((E)GSM or DCS)
Usage	The type assigned to the channel (Guard, TCH, BCCH)
Occurrences	The number of times the frequency is used within the selection made (Target+Interfering Cells selection)
Used By	Indication of source of occurrences: • Manual: it has been manually added (interfering cells selection) • ADCE: the channel is used by a defined neighbor of the target cell • Own cell: the channel is used by the Target Cell

Table 3 Fields Description of Frequency Tool result

Double clicking on a row will trigger a channel search. Hence a quick check can be done on the distance between the potential channel and the nearest co-channel and adjacencies.

2.2.2.1 DriveTest Tool

a. Finding and opening drivetest information

Opening the Drive Test Info tab reveals the methods to gather drivetest information into the mapping view.

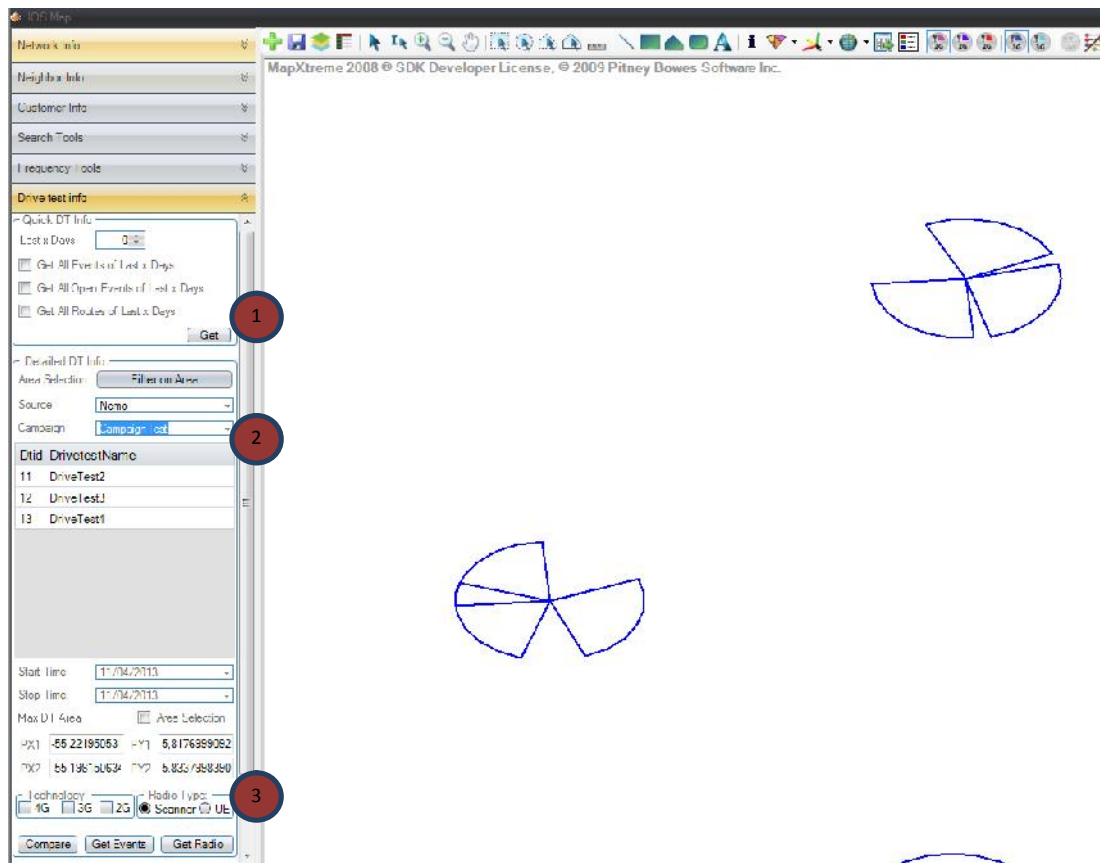


Figure 20 Drivetest Module - Explorer Bar

- 1 Quick DT info selection group enables to quickly call up the events and routes of the last X days.

By defining X in the Numeric UpDown controller, the user can get All events, All Open Events And/Or All Routes of the last X days. This covers all data in the database, and is hence **independent** of any datasource selection (source type (Nemo/Swissqual/...), Campaign, Technology..).

Selecting All Events or All Open events will automatically open the Drivetest Event form and populate it accordingly.

Fetching all routes will draw all parcels that have been touched during the last x days. This is a quick way to know where drivetests have been performed lately.

Detailed DT info:

Area Selection button, Source, Campaign and the grid itself are 4 filters to identify DriveTests of interest.

Two datasources are currently defined: Swissqual and Nemo. Each datasources is categorized in Campaigns and DrivetestIDs (DTID). Once a campaign selection is made, the listbox underneath is populated with drivetests.

Method 1: Filtering Using Source, Campaign and Grid

Step 1: Select Source. This will update the campaign field with all available campaigns for that source.

Step 2: Select Campaign: This will update the Grid with all available DriveTest for that particular campaign.

Step 3: Right click on header “DriveTestName” and choose the Sort or Filter capabilities.

Method 2: Filtering Using Area Selection

Step 1: Toggle the **Filter On Area** button. Now click and drag a rectangle on the map. This will set the area for which drivetest will be queried.

Step 2: Optionally set Source and Campaign

Step 3: Switch off the toggle button when satisfied with the results shown in the Grid

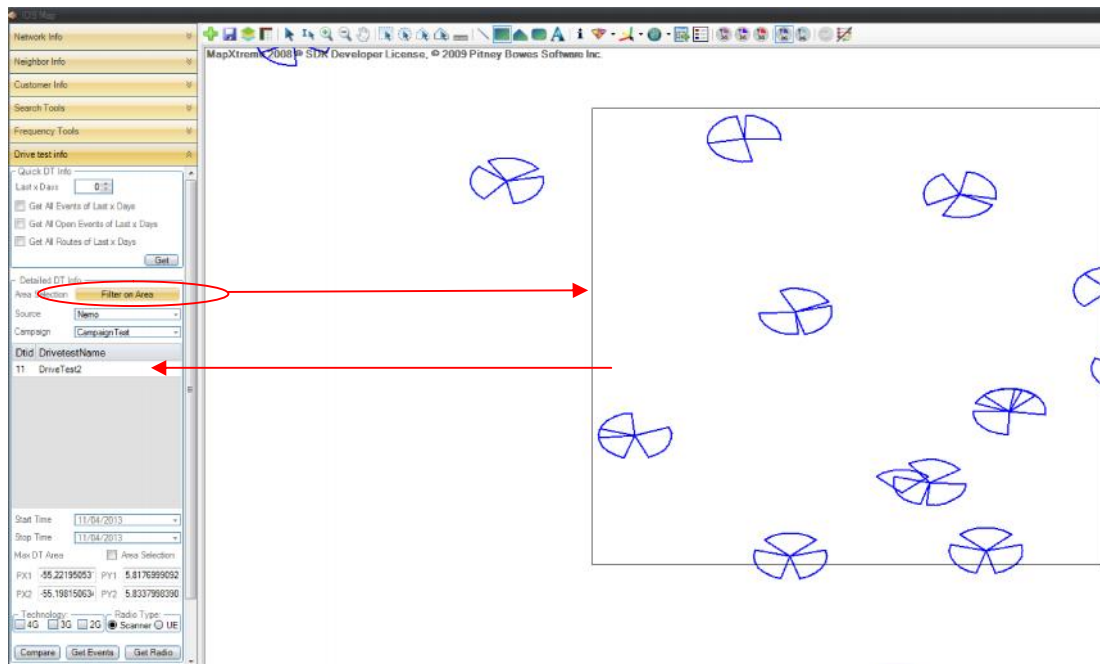


Figure 21 Drivetest Module - Example Filtering for Drivetests within an Area

After identifying the drivetest of interest take following steps to get the data from the drivetest server.

Step 1: select one or more drivetests from the Grid. This updates date filters to the minimum and maximum detected in that drivetest

Step 2: Filters:

- Optionally set start and end time filters for the data coming from the selected drivetests
- Optionally set a geographic filter, by clicking on the Area Selection checkbox and drawing a rectangle. Now only parcels falling within this rectangle for the selected drivetests will be downloaded.

Step 3: Select the technology of interest

3 Step 4: select the measurement device of interest (scanner/ue)

Step 5: Press “Get Radio” to get binned radio data for that particular drivetest. The data is presented in layers per technology and devicetype. This can be controlled in the LayerControl. Basic thematics are automatically generated. Additional thematics can be made by the user. All layers of the technology are made visible. Deactivate layers that are not of interest in the LayerControl.

Step 6: Press “Get Events” will open a new form showing the Failed events within that drivetest.

b. The Radio Data

Each parcel represents a collection of radio measurements averaged within approximately 32m x 32m per technology and per technology layer (UARFCN/EARFCN).

Default thematics are created per layer:

- GSM: RxLev/RxQual
- UMTS: RSCP/ECNO/PilotPollution
- LTE: RSRP/RSRQ

- 1 The layer control shows what drivetest layers are currently active and enables also the manipulation (making visible, adding/modifying themes, etc..) of the data.
- 3 The Parcel Selector tool in the Maptoolbar has two functions: the selection function makes it possible to show what the source of the signals are in that parcel or in a selection of parcels. The hover function is same, but in that mode you don't have to click on a parcel but just move the mouse over it.

Color Code for the source lines are for 3G:

- Green: serving cell
- Yellow: measured or scanned neighbor cell (or in active set)
- Red: pilot pollution (all signals within 6db of serving)

For more info, see 2.2.3.



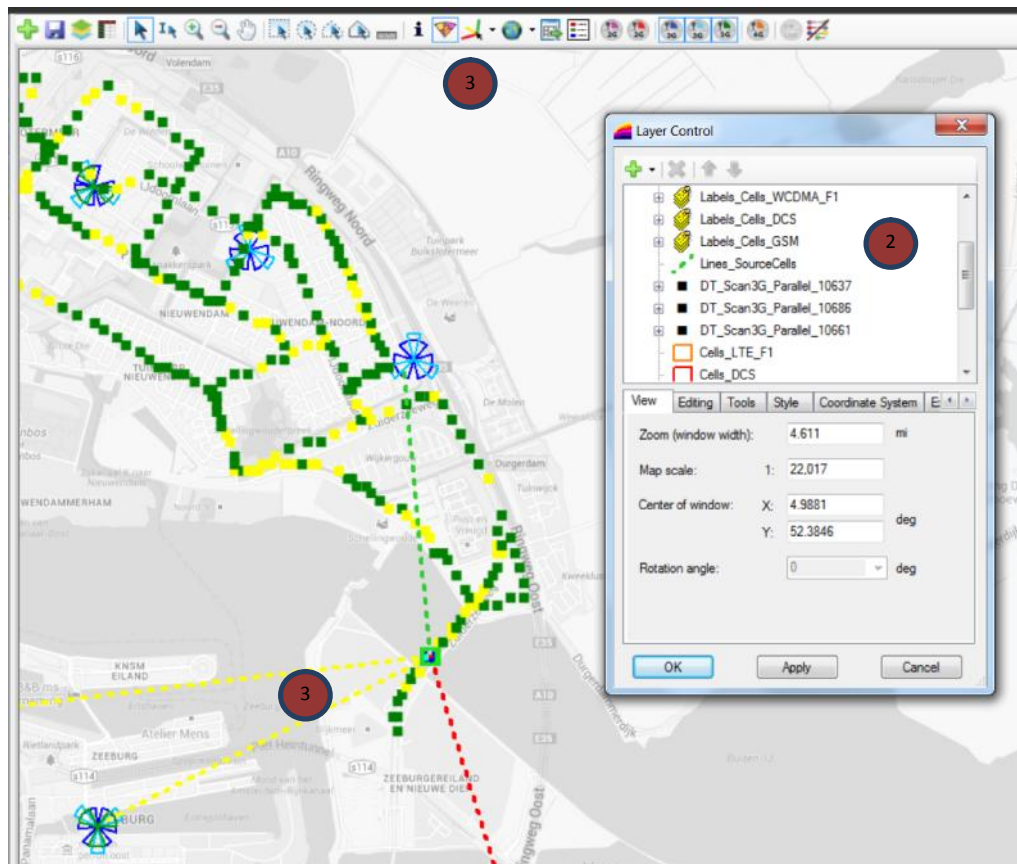


Figure 22 Drivetest Module - Parcel Selector

c. The Event Data

The event viewer represents all failed events (Failed/Dropped) for the selected drivetest. In contrast to the “Get Radio” data, the failed events data also holds raw radio data (unbinned) for this particular session, enabling detailed verification.

Three tabs are present:

- Analysis:

- 1 ○ The description box contains the analysis made by the engineer responsible for analyzing the drivetest data. The Comments box is reserved for reviewing engineers to add feedback to the responsible engineer. Event tracking and status information is shown at right side of form.
- 2 ○ Using the Update Status button, the engineer can update status or store feedback on the server

The screenshot displays the 'Drivetest Event Manager' application. At the top, a table lists event data with columns: EventID, DID, TimeStamp, Service, Technology, Failure Type, and LogFile. Below the table, the 'Analysis' tab is active, showing a form with a 'Description' field (containing 'Failed FTP DL: Big city: Amsterdam'), a 'Comments' field, and a right-hand panel with status-related dropdowns and an 'Update Status' button. Red circles with numbers 1 and 2 highlight the description/comments area and the update status button, respectively.

EventID	DID	TimeStamp	Service	Technology	Failure Type	LogFile
1	13050741	07-05-2013 01:34:12	da	umts	setup	
2	13050741	07-05-2013 11:53:58	da	umts	drop	
3	13050741	07-05-2013 12:54:40	da	umts	drop	
4	13050741	07-05-2013 01:31:20	da	umts	setup	
5	13050741	07-05-2013 03:44:58	da	umts	setup	
6	13050741	07-05-2013 09:54:41	da	umts	drop	

- Charts: The basic raw radio data of the selected event can be charted.
 - The chart shows basic radio data (RSCP, RSRP, ECNO, RXLEV) for the session selected in the event grid.
 - 1 ○ The radio can be shown per technology
 - 2 ○ The listbox of messages enable the user to show the message event on the chart. The Orange line indicates the timestamp of the failure event.
 - When “Map Event Data” is shown on the Map (see next paragraph), it is possible to click on the chart elements to highlight where this chart element is on the map.

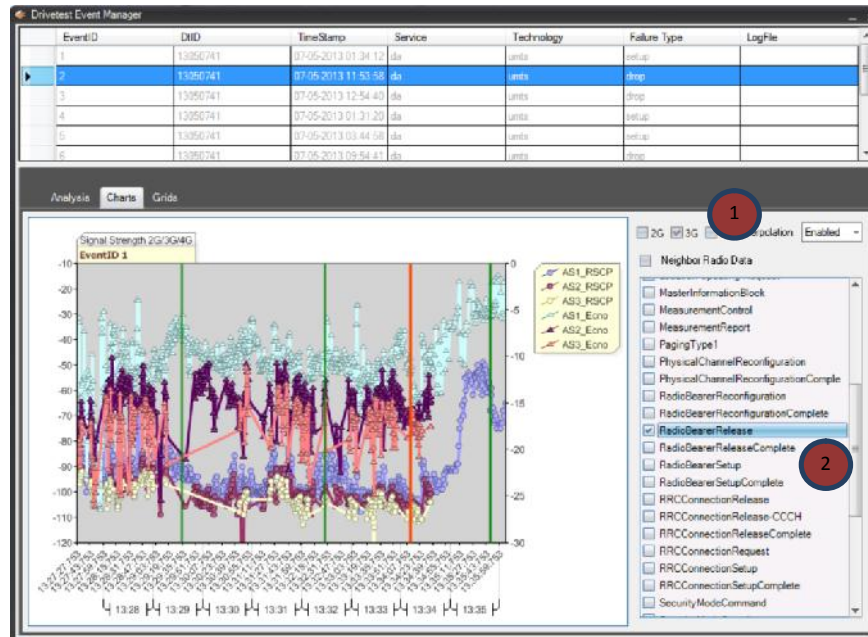


Figure 23 Drivetest Module - Chart Tab

- Grid: the raw data of the event is shown in a grid
 - o All raw data of the event that is available on the server can be called in the Grids tab.
 - o Selecting the technology filters the data on specific technology
 - o The Map Event button draws the radio data for all technologies on the map.
 - o Clicking on a row in the grid, highlights the selected data point on the map.
 - Thematics can be applied to the DT_EventGrid table that is created in the map view

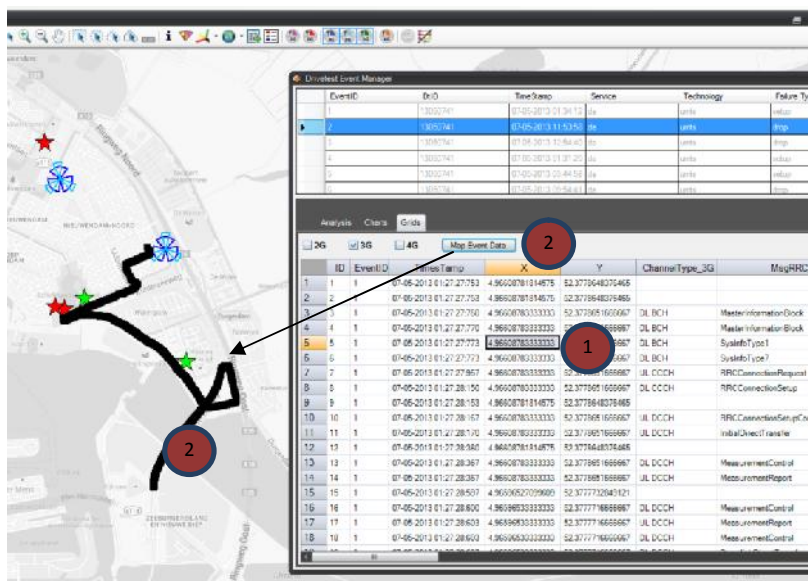


Figure 24 Drivetest Module - Event Manager - Grid&Mapping

d. CellFootPrint Function

- 1 The vast amount of data available on the server can be used to plot Cellfootprint information. All signals that were detected in all drivetest can be called for a specific cell. All parcels that have this cellid will be shown. For parcels that have overlap in time, the latest parcel will be shown.

This enables the visualization of scanner footprint and UE footprint and gives indication on real serving and NB coverage area of that cell.

The dropdown button also has the option to filter for *From Date* and *To Date*. This enables filtering the cellfootprint data, taking into account the time window set.

Multiple thematics are generated for this footprint per technology. This can be controlled using the LayerControl.

Mapinfo Layer Control:

When CellFootPrint is shown, the data is stored in a mapinfo layer called:

CF_<TECHNOLOGY>.

Method:

- Toggle the CellFootprint button to Scanner or UE
- Click any cell on the map (for which there's DT data).
- To Stop: set the CellFootPrint button back to "none". This will remove all tables and thematics related to CellFootprint.

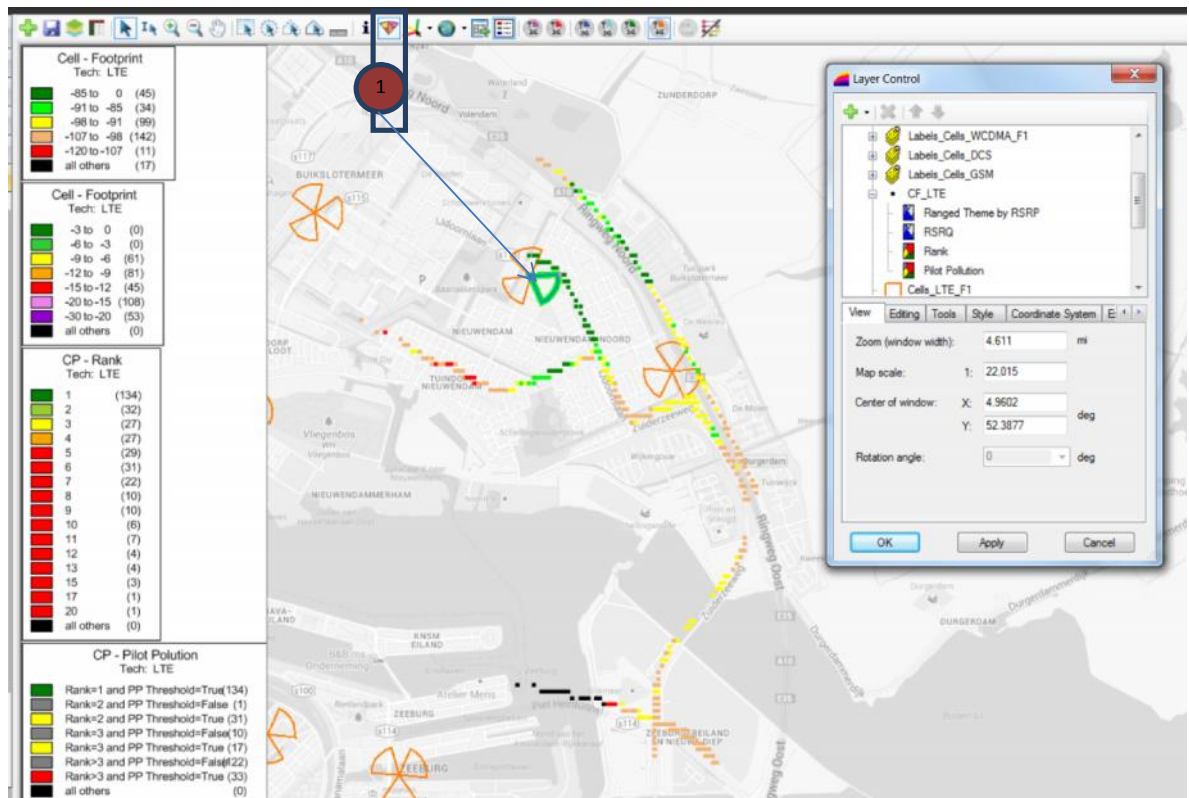


Figure 25 Drivetest Module - CellFootPrint

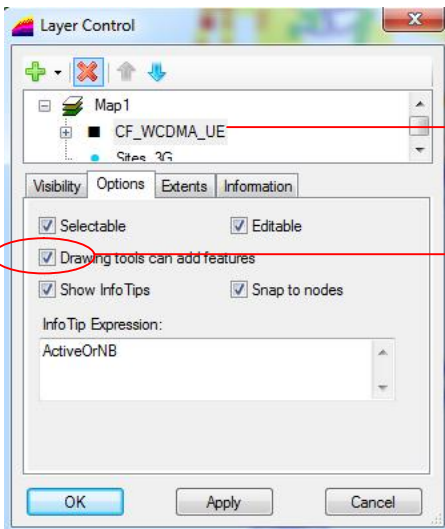
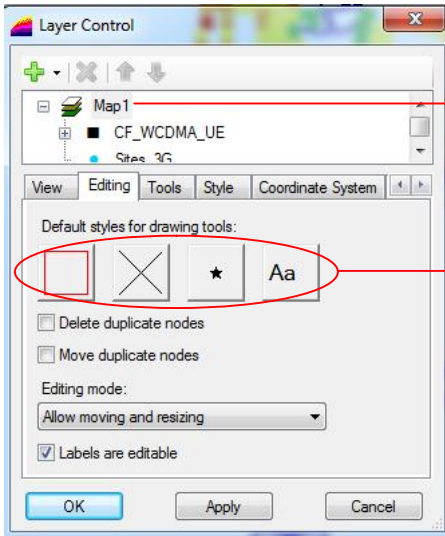
2.2.3 IOS – Map – MapToolbar



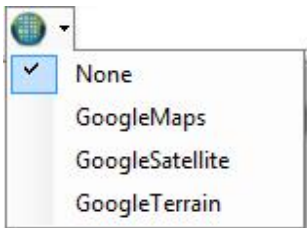
Figure 26 Maptoolbar

	Open Mapinfo .TAB file or IOS Workspace .MWS file
	Save a mapped table as a Mapinfo .TAB file
	Open the Layer Control
	Browse through a Mapinfo table. The table contents will be shown in the Map Grid.
	Map Selection Tool
	Map Layer Info Tool The tool enables to view the contents of all layers underneath the mouse icon. A click will open up the Selection Info window as shown in figure 14. The first list box displays all the layer-names that are beneath the mouse icon. Clicking on one of the entries will update the second listbox with the actual content of the object clicked in the selected layer. Layers under Mouse Content of Cells_WCDMA_F1 object clicked
	Zoom In, Zoom Out, Pan

Figure 27 Selection Info - Map Layer Info Tool

	Square selection, Circular selection, free hand selection, Inside Polygon selection
	Distance Measurement Tool (Toggle Button) If button is activated, the mouse cursor can be used to measure distances. Clicking and moving the mouse to set waypoints will show the distances in the status bar of the map window. Escape or Double click will remove the measurement lines.
	Choose layer to add the drawing:  Select Layer to add drawing Check highlighted box to activate layer for adding drawings Set Default Drawing properties in Layer Control:  Select Top Layer Item 4 style modifiers are available: - Region (rectangle) - Line - Point - Text

	A quick button to bring up the Selection Info Window
	Cell Footprint (Toggle Button) The toggle button has 3 states: • None: the cellfootprint function is deactivated • Scanner: the cellfootprint of scanner data will be shown • UE: the cellfootprint of UE data will be shown If button is activated, then the selection of a cell will result in a query towards the Drivetest server to get the latest radio measurements of that cell. All <u>latest</u> parcels where this cellid+lac was detected will be shown. To control the time window of the cellfootprint, the 4th option in the dropdown menu can be used. Setting the From Date and To Date datetimepickers enables time filtering for the cellfootprint data. By default these datetimepickers are set to minimum available date of all drivetests to today. Thematics: Some default thematics have been defined: • 2G: RXLEV, RXQUAL • 3G: RSCP, ECNO, RANK • 4G: RSRP, RSRQ, RANK Changes made to these thematics will be remember on the local machine. Mapinfo Tables: Every time a cell is clicked, a table named CF_<Technology> is created and mapped.
	Parcel Selector (Toggle Button) The toggle button has 3 states: • None: the parcel selector is deactivated • Parcel Hover: when this is active, the user can just move his move over a parcel to identify the source cells • Parcel Select: when this is active, the user can select one or more parcels using the normal selection tools to

	identify the source cells. Color Coding of the lines: 2G Scanner: • Green: Strongest Cell based on RxLev • Yellow: 2e or worse cell based on Rxlev 2G UE: • Green: Serving Cell • Black: Detected NB Cell 3G Scanner: • Green: Serving Cell • Yellow: 2e and 3e strongest cell within 6dB of serving cell • Red: 4e strongest cell within 6dB of serving cell (pilot pollution) 3G UE: • Green: Serving Cell • Yellow: Cell in Active Set • Black: Detected NB Cell 4G Scanner: 4G UE:
	Background Map Source Selection Four option are given: • None: No background map. The preferred option if a local Mapinfo table is used. • GoogleMaps: background will be set as semi transparent, greyed map based on Google Maps service • GoogleSatellite: background will be set as opaque, full color satellite imagery based on Google Maps service • GoogleTerrain: background will be set as opaque, full color terrain data with terrain elevation information
	Show the Map Grid

	Show the legend.
	Technology Layer Quick Display buttons When activated, all layers starting with naming “Cells_<Tech>” will be displayed. When de-activated, all layers starting with naming “Cells_<Tech>” will be hidden. These buttons are dynamically configured at startup and specific to a customer deployment. Example: Magenta button: 2G – EGSM, Layer name: Cells_EGSM Red button: 2G – DCS, Layer name: Cells_DCS Blue button: 3G – WCDMA F1, Layer name: Cells_WCDMA_F1 Cyan button: 3G – WCDMA F2, Layer name: Cells_WCDMA_F2
	Show/Hide Network State and Tickets layers This button is enabled if Network State information is available through pressing the Network State button in the “Network Info – Explorer Bar Item”. If activated, all network layers ending with “_Status” or “_Tickets” will be shown. If deactivated, all network layers ending with “_Status” or “_Tickets” will be hidden.
	Removes all the layers except the Cells_<Tech> layers.

2.2.4 IOS – Map – Selection Window

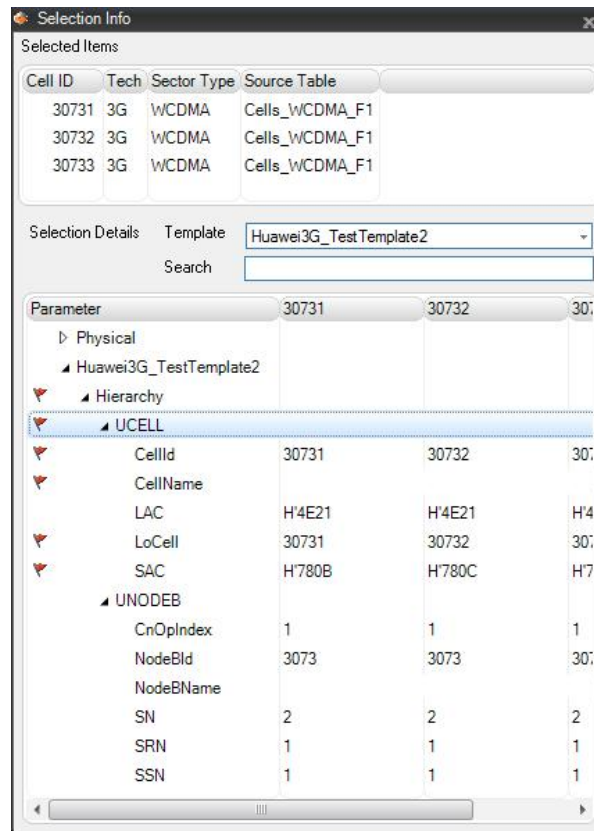


Figure 28 Selection Window

Although visually similar to the Info Window, the Selection Window is specifically designed to read cell based information. The top listbox provides the information regarding the object technology and layer. The bottom treeview has two distinct components:

- Physical: the information that is retrieved from the network configuration layer, stored on the IOS server and updated on a daily basis
- Templates: the **CM** cell parameter information based on live OSS or CM datawarehouse and the selection made in the Template dropdown box. A preconfigured set has been defined in IOS. Templates can be managed in the Manager – CM (see chapter 2.2.6).

The Selection Window is dynamic in such that multiple selection are visualized instantly and compared on the fly. A delta between the first entry (first column) and the other entries (other columns) is flagged by a red flag on the left side.

A search function enables quick searching through the left column. Typing any text and hit enter to get the next match. The result will be focused upon.

As figure 20 shows, a context menu is available at parameter level, providing:

- Quick access to detailed description of the parameter highlighted (if available)
- Quick mapping of the parameter highlighted over all objects in the map window. An example is shown for MaxTxPower.

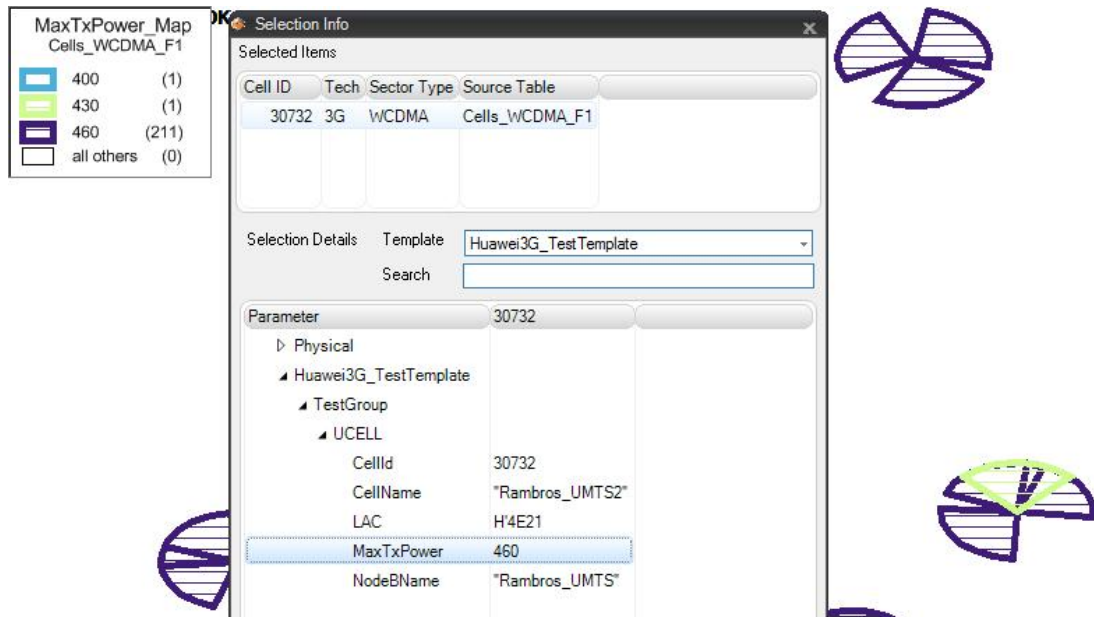


Figure 29 Selection Window - Context Menu - Parameter Map

2.2.5 IOS – Map – Context Menu

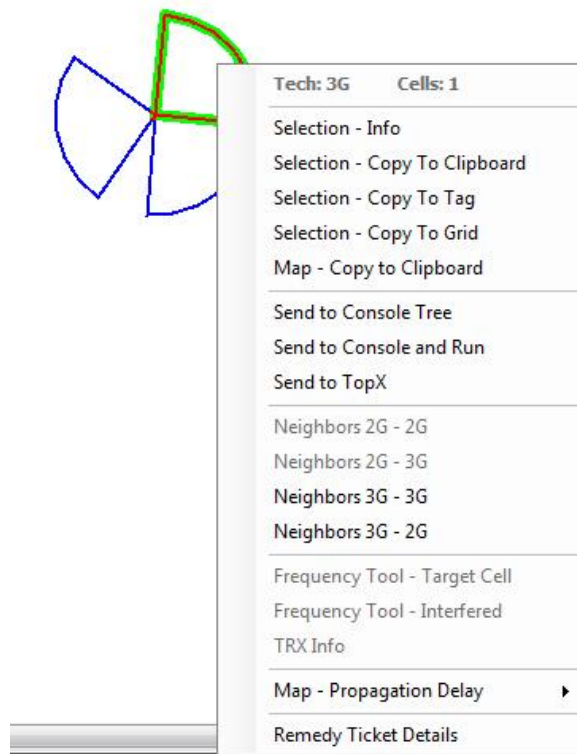
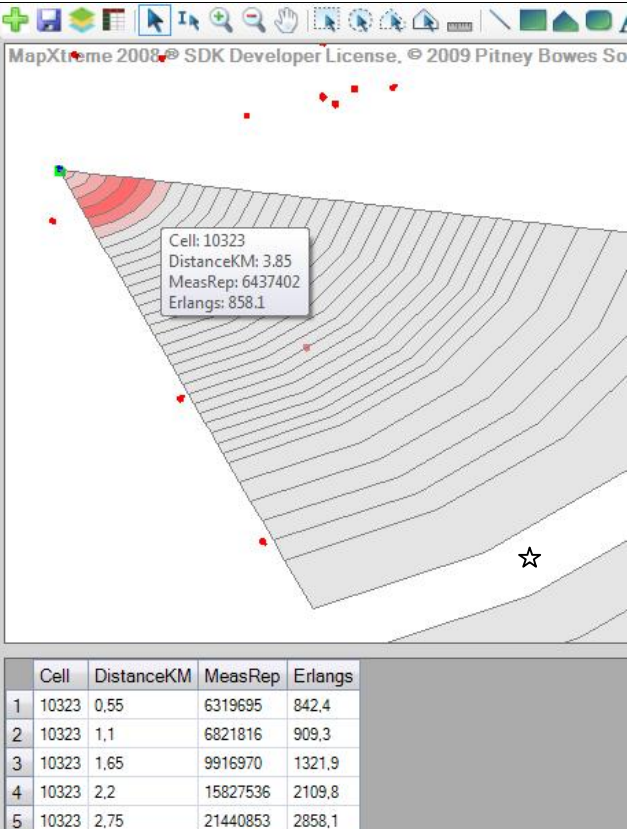


Figure 30 Map Context Menu

Tech: 3G Cells: 1	Tech indicates what technology has been selected. Cells indicate how many cells for that technology are selected on the map.
Selection - Info	Launches the Selection Window (2.2.4), with the selection as startup.
Selection - Copy To Clipboard	Copies the physical information of the selection into the clipboard
Selection - Copy To Tag	Copies the CELLID information to a TAG. This function only works if the user has created a Tag in the Tag Manager.
Selection - Copy To Grid	Copies the physical information to the grid on the map
Map - Copy to Clipboard	Copies a screenshot of the the map view to the clipboard.
Send to Console Tree	The selection on the map is synchronized to the appropriate object tree in the Console Window.

Send to Console and Run	The selection on the map is synchronized to the appropriate object tree in the Console Window. The statistics are started
Send to TopX	The selection on the map is synchronized to the appropriate object tree in the Console Window.
Neighbors 2G - 2G Neighbors 2G - 3G Neighbors 3G - 3G Neighbors 3G - 2G	This is an alternative way to query for neighbor relation definitions and statistics. The result is the same as described in 2.2.2
Frequency Tool - Target Cell Frequency Tool - Interfered	This is an alternative way to use the Frequency Tool. Usage: Step 1: Select the cell for which you want to find a free frequency Step 2: right click and choose Target Cell Step 3: select a cluster of cells surrounding that cell (the target cell can be included), that you are considering as potential interferers. Go to Frequency Tool explorer bar and hit Calculate!
TRX Info	If a 2G Cell is selected, this will query for the TRX info for that particular cell, showing one line per TRX and the channel configuration for that TRX. (If available).
Map - Propagation Delay ▶ Map - Timing Advance ▶	For 3G: A Traffic distributed over Propagation Delay can be drawn on the map, giving the user good insights on where the traffic is coming from. For 2G: The traffic distributed over Timing Advance slots is drawn on the map.

	<table><tr><th></th><th>Cell</th><th>DistanceKM</th><th>MeasRep</th><th>Erlangs</th></tr><tr><td>1</td><td>10323</td><td>0,55</td><td>6319695</td><td>842,4</td></tr><tr><td>2</td><td>10323</td><td>1,1</td><td>6821816</td><td>909,3</td></tr><tr><td>3</td><td>10323</td><td>1,65</td><td>9916970</td><td>1321,9</td></tr><tr><td>4</td><td>10323</td><td>2,2</td><td>15827536</td><td>2109,8</td></tr><tr><td>5</td><td>10323</td><td>2,75</td><td>21440853</td><td>2858,1</td></tr></table>		Cell	DistanceKM	MeasRep	Erlangs	1	10323	0,55	6319695	842,4	2	10323	1,1	6821816	909,3	3	10323	1,65	9916970	1321,9	4	10323	2,2	15827536	2109,8	5	10323	2,75	21440853	2858,1
	Cell	DistanceKM	MeasRep	Erlangs																											
1	10323	0,55	6319695	842,4																											
2	10323	1,1	6821816	909,3																											
3	10323	1,65	9916970	1321,9																											
4	10323	2,2	15827536	2109,8																											
5	10323	2,75	21440853	2858,1																											
	Figure 31 Timing Advance Sample The tooltip shows what can be seen in the grid: CellID, Distance, Number of Measurement Reports captured for that TA slot, and the traffic estimate based on these measurement reports. The default thematic, applied to each cell individually, is: • Red: maximum traffic value of the query result • Grey: minimum traffic value of the query result • No result: nothing will be drawn (see ☆)																														
Remedy Ticket Details	If a ticket interface is present, this function will launch a browser, requesting ticket information from the Remedy server.																														

2.2.6 Manager – Reports

The report editor is a tool that facilitates the creation of standard Microsoft PowerPoint based reports. It allows users to create and share reports containing charts. The reports are stored on the IOS Server.

The prerequisites for this function are:

1. a “Template.potx” file is present in the IOS program folder, or on the location that is defined in the IOS.exe.Config under the key: “ReportingTemplateFileName”
2. The Template.potX must have on its first page, 2 textboxes: 1 containing “<Title>” and 1 containing “<Subtitle>”. These are TAGS that will be replaced by the Report Manager during runtime.
3. A report group exists and the user is a member of that report group. Contact your administrator if this needs to be implemented

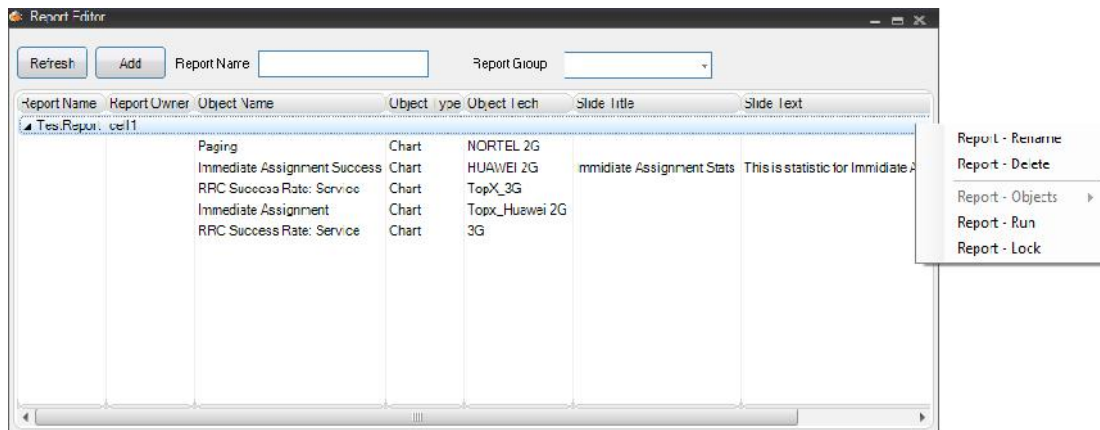


Figure 32 Report Editor

Workflow to create a report:

- Enter the report name in the text box, select the Report Group to which the report needs to be added and click add. If no Report Group is available, contact your administrator. The report name will be stored together with the username on the IOS Server. The report will now be visible by any user, but only modifiable by the user whom created it.

- To add a chart, go to any chart on the Console Window, right click and “Report - Add chart” and choose the last report name that you have added. You know have added the selected chart into your report.
- Go back to the Report Window and click the triangle next to the report name. This will expand the report and describe the contents of the report.
 - o Right clicking on the charts gives the option to add, remove or change the position of the chart in the report.
 - o A custom title can be added by entering text into the textbox that pops up when the mouse is held over Edit Slide Text. By default, an empty title will show the Object Name as title in the end report.
 - o Additional text can be added to the slide by holding mouse over “Edit Slide Text”. Any text can be entered in the textbox that pops up. To save it: hit ctrl-Enter

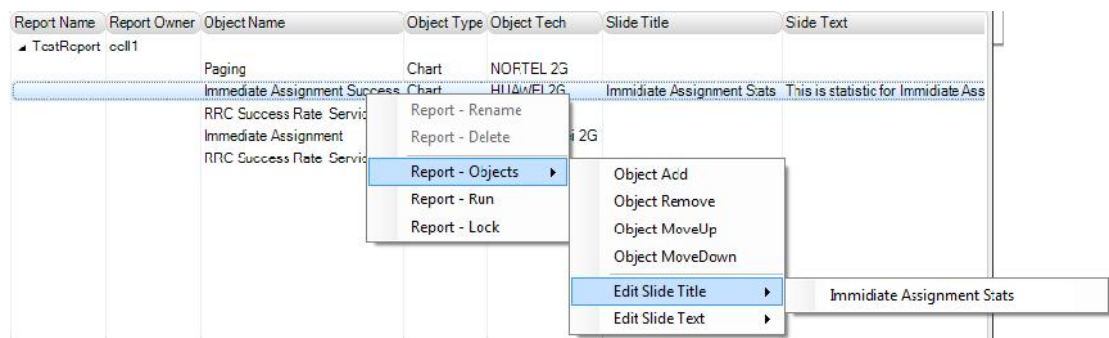


Figure 33 Report Manager - Context Menu

- To Create the PowerPoint report, right click on the *report name* next to the triangle, and choose Run. The report will be generated and stored in user data folder “C:\Users\<username>\AppData\Roaming\CellSens\IOS\<deploymentname>\data”

Workflow to modify a report:

- Expand a report by clicking on the triangle next to the report name
- Right clicking on an object of the report allows you to either Remove the object, Move the object up in the order or Move the object down, or, change the title and slide text. This action is only allowed by the owner of the report or if the report is “unlocked” by the owner.

Workflow to delete a report:

- Right click on a report name and select delete. This action is only allowed by the owner of the report.

Report - Lock

By default, when a user adds a new report into the server, the report can't be modified by other users. The lock is "enabled".

When the owner changes the lock to "disabled", any user that is member of the report group can make changes to that particular report.

2.2.7 Manager – Tags

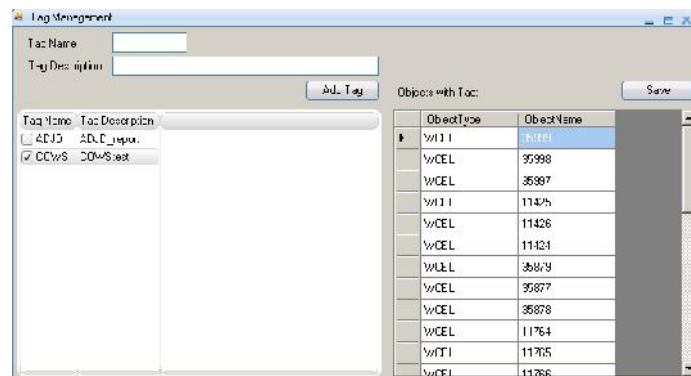


Figure 34 Tag Management Window

Tag Management allows a specific user to create its own object sets by assigning a tag to an object.

Workflow to Create a Tag:

- Enter Tag Name and press Add button to store the name on the IOS Server
- Click on the Tag name in the tree. On the right side, an empty grid will be displayed.
- Fill the grid with the desired objects using following methods:
 - Manual:
 - Enter object type, currently WCEL or CEL in the first column
 - Enter Cell_ID, of the object you wish to add
 - Copy & Paste from Excel
 - In Excel, the first column should hold the objecttype, the second column the cell id. Copy the entire table, excluding the headers.
 - Select the first row, right click and select Paste.
 - From Object Tree:
 - Make a selection in the object tree
 - Right click and select “Copy To Tag”
 - Select the tag to which you want to add the objects
 - From Map:
 - Select the cells of interest
 - Right click and select “Copy To Tag”
 - Select the tag to which you want to add the object
- Press the save button to commit the table to the IOS Server
- The TAG is now available for all users in all object trees in the Console Window.

Workflow to Modify a Tag:

- Only owners of the Tag are able to Modify the tree
- Select the tag in the treeview of the Tag Management Window
- Remove or add records in the grid next to the tree
- Press save

Workflow to Delete a Tag:

- Select the tag in the treeview of the Tag Management Window
- Right-click on the tag and select delete to update the IOS Server

2.2.8 Manager – CM

The Manager-CM module is a separate form that enables the creation and managing of CM (configuration management) templates. Templates have currently 2 key functions:

- The Selection Window uses templates to show what CM data needs to be queried for when the user selects a cell on the map
- The template can be used in the SON environment for parameter reference checks.

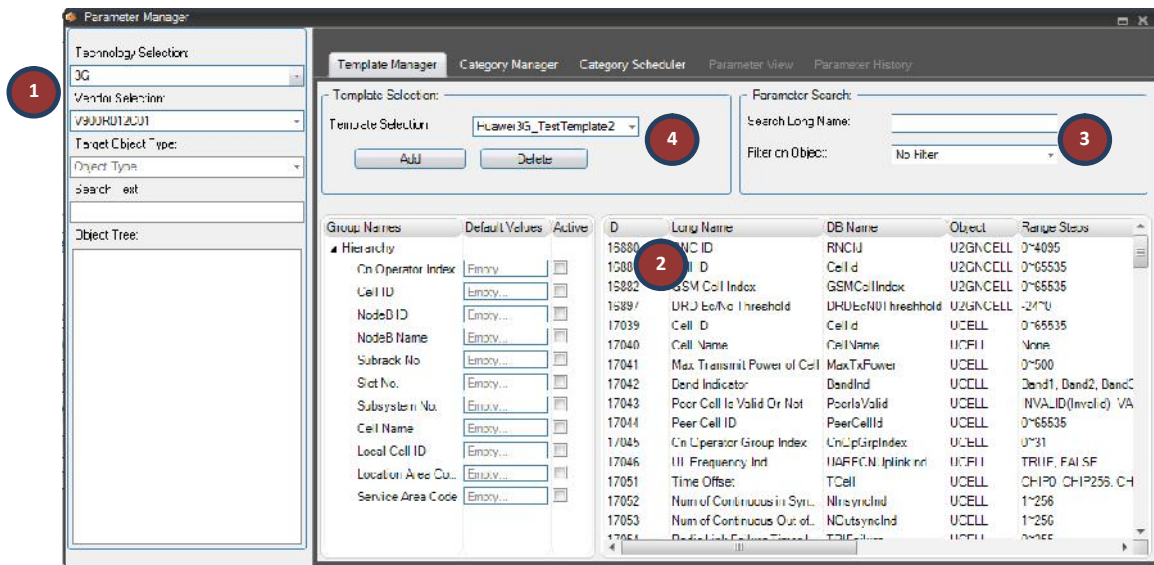


Figure 35 CM Manager

Workflow to Create a New Template:

1. Select the technology and vendor of interest. When selection of vendor is completed, the application will query the database for the relevant CM parameters. The result is shown in the list on the right hand side of the form
2. Click Add Template:
 - o A new dialog box pops up which requires entering the name for the template
 - o When hitting ok, the name will be saved to the database, along with the username that has created it. From here on, only the user that created the Template will be able to modify or delete it.
4. To organize the parameters in the template, groups can be created.
 - o Right click on the treeview under the Add button and choose Add New Group
 - o A textbox is presented in which you can provide a name for the new group

3

- To add a parameter to the group, drag and drop from the right list of parameters to the groupname. To easily find the parameters of interest there are 2 filter mechanism:
 - o The long name textbox filter enables the user to wildcard search any text within the longname textbox.
 - o The filter on object can be used if the user is interested in a particular objecttype.
- Three columns are presented once the parameter has been added: the name of the parameter, a field where the user can optionally enter a default value, and, a checkbox where the user can enable a parameter to be part of the (optional) SON reference check. The latter two are only applicable if the optional SON reference check is active.

Workflow to Modify an existing template:

- Select Technology and Vendor of interest
- Select the Template you wish to modify. Only creators can modify templates
- All modifiers are available through right clicking on the parameter template list:

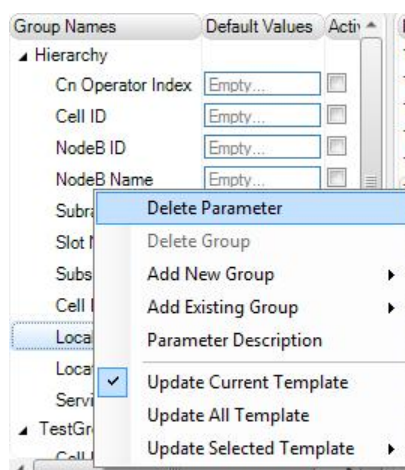


Figure 36 CM Manager - Context Menu

- o Delete Parameter: when a parameter is selected, this function will remove the parameter from the template
- o Delete group: when a group name is selected, this function will remove the group from the template
- o Add New Group: will present a textbox to enter new group name
- o Add Existing Group: will present with a dropdown box showing all existing groups defined on the IOS server. By clicking on one of these, the whole group will be copied to your template.

- o Parameter Description: will pop-up a new form with descriptions of the parameter selected. This function can also be called from the parameter list on the right side of the form
- o The 3 Update menu items relate to Default value setting and the Active setting:
 - When Update Current Template is checked: changes made to the default value and active checkbox will only apply to the template that is currently being worked upon
 - When Update All Template is checked: changes made to the default value and active checkbox will propagate to the same parameter in ALL templates
 - When Update Selected Template is checked: changes made to the default value and active checkbox will only apply to the selected template in the dropdown box presented.

2.2.9 Manager – PM KPI

The module is presented in a new form and its purpose is to create and manage the performance management KPIs that are available in the Console Window.

Only users with elevated permissions are allowed in this module.

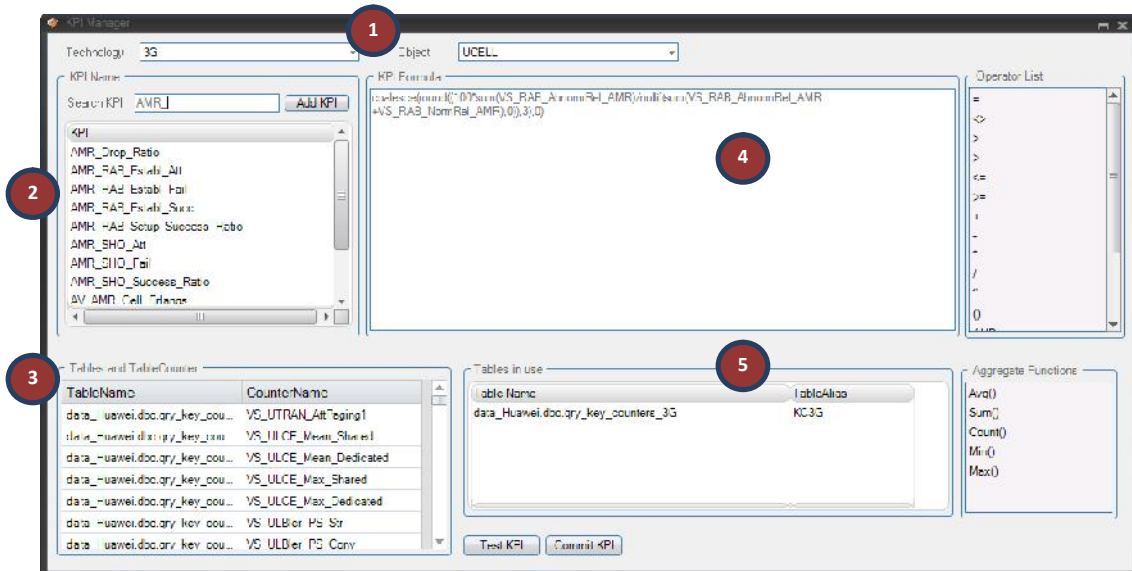


Figure 37 KPI Manager

Workflow to Create a New KPI:

- 1 • Select technology and object for which a new KPI needs to be created
- 2 ○ The existing KPI list to be updated
- 3 ○ The Tables and TableCounter grid will be updated
- Hit Add KPI: a dialog box will pop up in which the user can enter the KPI name. After confirming the name, the KPI list will select and focus on the new entry. Now the KPI Formula can be created.

Build a formula:

- 4 **PREREQUISITES**
 - Formulas are based on counters in certain measurement tables. These are presented in the Tables and TableCounter grid. Each technology and object has its own counter list. To find a counter, the filter and sort methods available in the right click menu of the grids header can be used.
 - A Formula must always be an aggregation. This means that I should work in a query for 1 cell or for multiple cells. As a result, every counter must be part of a aggregating function like SUM, AVG, etc..
 - A formula represents a field in an SQL statement. Hence the formula must be 100% SQL compliant.

PROCESS

Example 1:

- o The target formula we want to implement in this example is:

$$CS_{DropRatio} = 100 * \frac{VS_RAB_AbnormRel_CS}{VS_RAB_AbnormRel_CS + VS_RAB_NormRel_CS}$$

- o Step 1: identify which aggregators to use. In this case all counters are cumulative integer numbers. Hence SUM() aggregation can be used on nominator and denominator.
- o Step 2: type: "100 * " and drag and drop SUM() from the Aggregate functions
- o Step 3: find the counter: VS_RAB_AbnormRel_CS by right clicking on the header of the countertable grid and choosing "Filter". Set Show Rows where to "Contains" and type as Value: "normRel". The filter is now applied and grid shows only counters that contain the string "normRel". Find the counter VS_RAB_AbnormRel_CS and drag and drop this onto the KPI formula text box.
 - The counter is automatically added between the brackets of the aggregate function.
 - The counter consists of 2 elements: the "tablename"."countername".
 - The Tables in use list is updated to indicate what tables are currently used in the formula.
- o Step 4: drag and drop the Divisor (or press "/")
 - This will extend the existing text with:
 - / NULLIF(<enter denominotar here> ,0) for MSSQL based system
 - / NVL(<enter denominotar here>,0) for Oracle based system
 - This auto-text is meant to protect SQL query from "Divide by zero". Don't remove this text.
- o Step 5: add another aggregation function SUM() for the denominator.
- o Step 6: add the 2 counters separated with the + sign
- o Step 7: If it is desired to have a rounded result, the round(,) function can be used. Encapsulate the existing formula with : "Round(<formula>,x)" with x being the number of decimal integers desired.
- o The end result should look like this:



KPI Formula

Round(100 * Sum(KC3G.VS_RAB_AbnormRel_CS) / NULLIF(Sum(KC3G.VS_RAB_AbnormRel_CS+ KC3G.VS_RAB_NormRel_CS),0),2)

- o Step 8: Test KPI button can be used to test the KPI in a real query. The response of the query can be used to debug the formula
- o Step 9: Commit the KPI: This will test and write the KPI to the database. From then on the KPI is available to be added to a chart.

REMARK:

On Commit, the KPI formula text will be encapsulated by a NULL to 0 protection, represented by the text:

- ISNULL(<formula>,0) for MSSQL
- COALESCE(<formula>,0) for Oracle

When modifying a KPI, this encapsulation should be preserved !

Workflow to Modify a KPI:

- Select technology and object for which a new KPI needs to be created
 - o The existing KPI list to be updated
 - o The Tables and TableCounter grid will be updated
- Search the kpi to be modified using the search text box.
- To Rename the KPI: right click on the KPI and hit rename
- To Remove the KPI: right click on the KPI and hit Remove
- If the formula needs changing it is recommended to always use drag and drop operations to ensure the syntax for table.countername is correct. Always Test And Commit to save the updated KPI

REMARK:

When new measurements are made available by the network vendor, and these are not available in the IOS solution, then contact your administrator.

2.2.10 Manager – Charts

Create and managing charts is handled within the Chart Manager. There are two types of permissions to Chart Managers:

- Normal user will have access to Chart manager, only to create its own Custom Charts and the Chartsetnames combobox is greyed out/disabled.
- Elevated user will have access to Chart manager, to create its own Custom Charts but also to modify charts of the global chartset, visible to anyone. Ask the administrator if elevated permissions are required. The Chartsetnames combobox is enabled.

Before starting to design or modify charts, it is recommended to have sample data to work with. The Chart Manager uses “in – memory” datatables of queries that have been fired in the last time. To know if these are available, the Chart Manager checks for this at the loading of the form.

Per Technology there are 2 states:

- 1 • Data for <tech> Charts – Not available => it is recommended to run a sample query in the console window.
- Data for <tech> Charts is available. No action required.

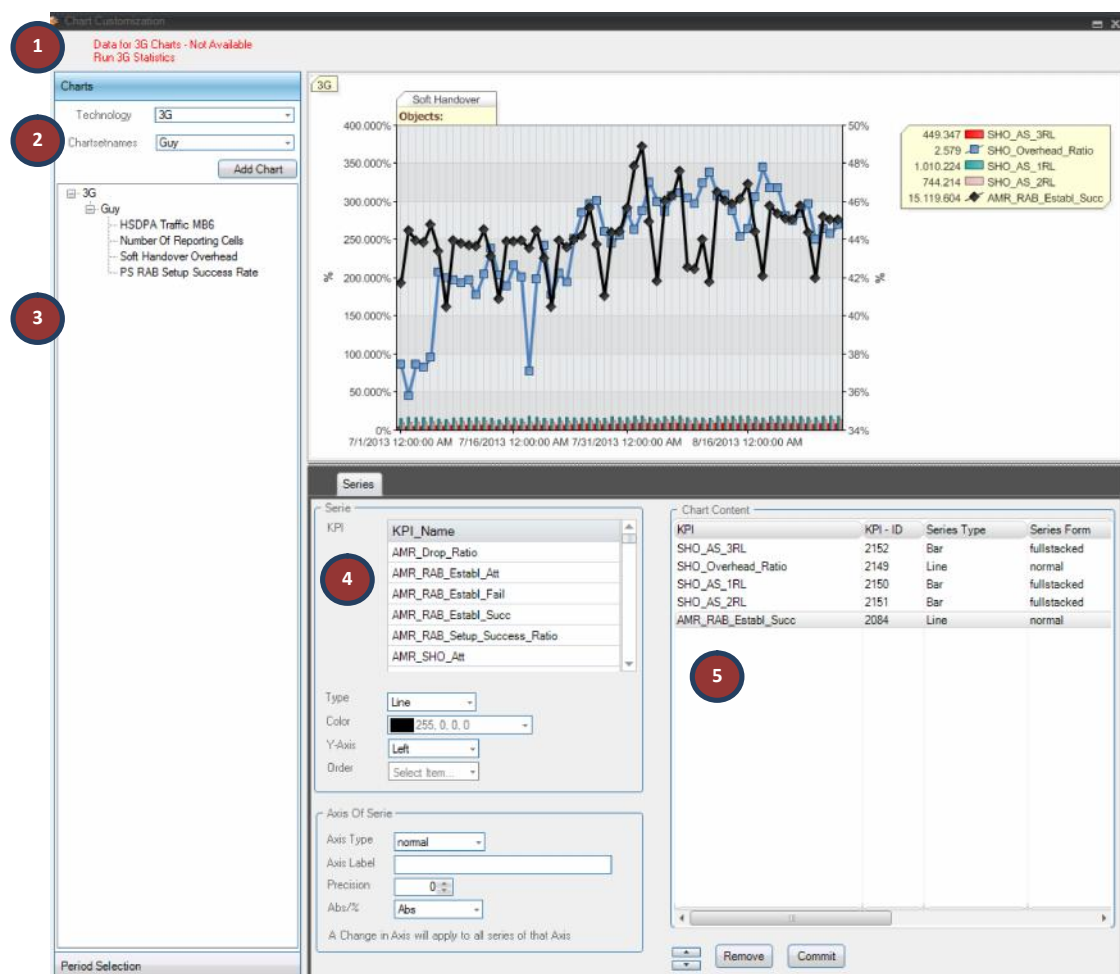


Figure 38 Chart Manager

Workflow to Create a New Chart in Custom Tab:

Method 1: Create a New Chart from scratch

- Step 1: Select Technology.
 - By default, the ChartSetName of the user will be set and the treeview containing the charts will be updated
 - The KPI grid updates to KPIs for the selected technology
- Step 2: Right click under the user name (or category name) and select “Chart - Add Empty”. The Tree with charts is refreshed and a new chart with title “Add Title” is added
- Step 3: Click on the “Add Title” to rename it.
- Step 4: Find and select the KPI of interest. Finding a KPI can be achieved using the filter capabilities of the grid by right clicking on the header
- Step 5: Drag and Drop from the Grid to the List containing the series on the right side of the form. This should add a new entry with all the settings as set.
- Step 6: Set the chart series properties of the KPI. Make sure you have selected the correct KPI in the Chart Content list. Any change that is made in the Series controls is reflected on the selected KPI in the Chart Content list.
 - Type: can be either line or bar
 - Y-axis: assigns the KPI to the left or right axis
 - Order: defines the sort order for TopX chart data. This is only enabled when TopX technology is selected.
 - Axis type can be 3 types:
 - Normal
 - Stacked: used for Bar typed series. The results of each bar series is stacked ontop of eachother. See Example

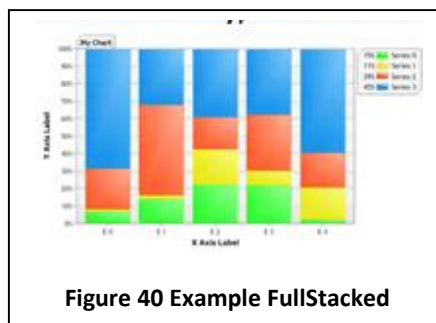


Figure 40 Example FullStacked

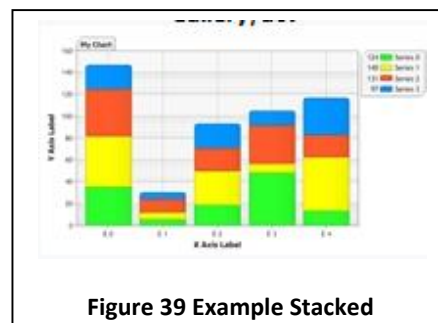


Figure 39 Example Stacked

- FullStacked: used for Bar typed series. The result of each bar series is stacked ontop of eachother and normalized. See Example

- Axis Label: Enter a label for the Y-axis
 - Precision: defines how many digits in the decimal should be shown on the axis
 - Abs/%: Defines if axis is containing a value that is a percentage (e.g. a ratio) or is containing an absolute value (e.g. number of attempts).
- Repeat Step 4, 5 and 6 to add more series.
- Step 7: Hit Commit. This will write the result to the database, and redraw the sample chart above the series tab.

The chart will now be visible in the Custom tab when the user re-runs the statistics for the technology that a chart has been made for.

Method 2: Create a New Chart based on existing Chart

- Step 1: Choose the chart of interest in the Console Window and right click -> add to custom. This action writes the chart configuration to your personal Custom tab.
- Step 2: Open the Chart Manager for and select the technology that relates to the chart you have added to your Custom Tab.
 - The tree is updated and should contain the name of the chart that you have right clicked in step 1.
- Step 3: Select the chart in the tree. Now the chart will be drawn and the Chart Content will be filled. From here on, follow steps 6 and onwards from Method 1.

Modifying Charts:

- Renaming a Chart can be done by clicking on the chartname in the treeview.
- Changing the position of the chart can be achieved by dragging and dropping the chartname in the treeview.
- Deleting can be achieved by right clicking on it and selecting Delete.

REMARK:

When changing anything in the chart series, a Commit is required to store the change. All other changes like renames, positions and deletions do not require a Commit.

2.3 The IOS Console Window

The second core window of the IOS Smart Client is the “IOS Console Window” which is specialized in retrieving and visualizing network performance information.

The form is built up from 8 main Tabs as shown in Figure below



Figure 41 IOS Console Tabs

1 Dashboard: Tab providing overview of the performance of the network

2G: Tab specialized performance data for 2G

3G: Tab specialized performance data for 3G

4G: Tab specialized performance data for 4G

2 Each Technology Tab has subtabs covering Timebased and Objectbased statistics for the objects of interest.

Transmission: Tab specialized in time based performance data for Transmission related objects

Core: Tab specialized in Core specific time based performance data. Examples are MSS, SGSN, GGSN, MGW.

Trace: Tab specialized in visualizing trace information stored on the OSS.

SON: Tab containing reporting modules for semi or fully automated functions.

2.3.1 Dashboard

The Dashboard is the front GUI of the IOS Analytics module which is focused on providing the user with problem cells that matter to the organization. The IOS tickets are created automatically on a daily basis at night. All cells active in the network are subject to a set of performance rules. If a rule is breached, the cell will become a problem cell and will be assigned an IOS Ticket.

To distinguish the importance of each IOS Ticket, a score is assigned to each ticket. The score depends on the value of the rules breached. For example, 1 rule: if a cell has more than 10 RRC CS CALL SETUP FAILURES and if the cell has a CSSR < 90%, then the cell will get a score of 1000.

The rules are set currently by the administrator of the tool. It is recommended that rules are set together with the department using the tool.

Once an IOS Ticket is created, it receives the Status “Open”. It is expected that the responsible for the particular Area is assigning the ticket to himself and follows up the problem until the ticket reaches “Closed” status.

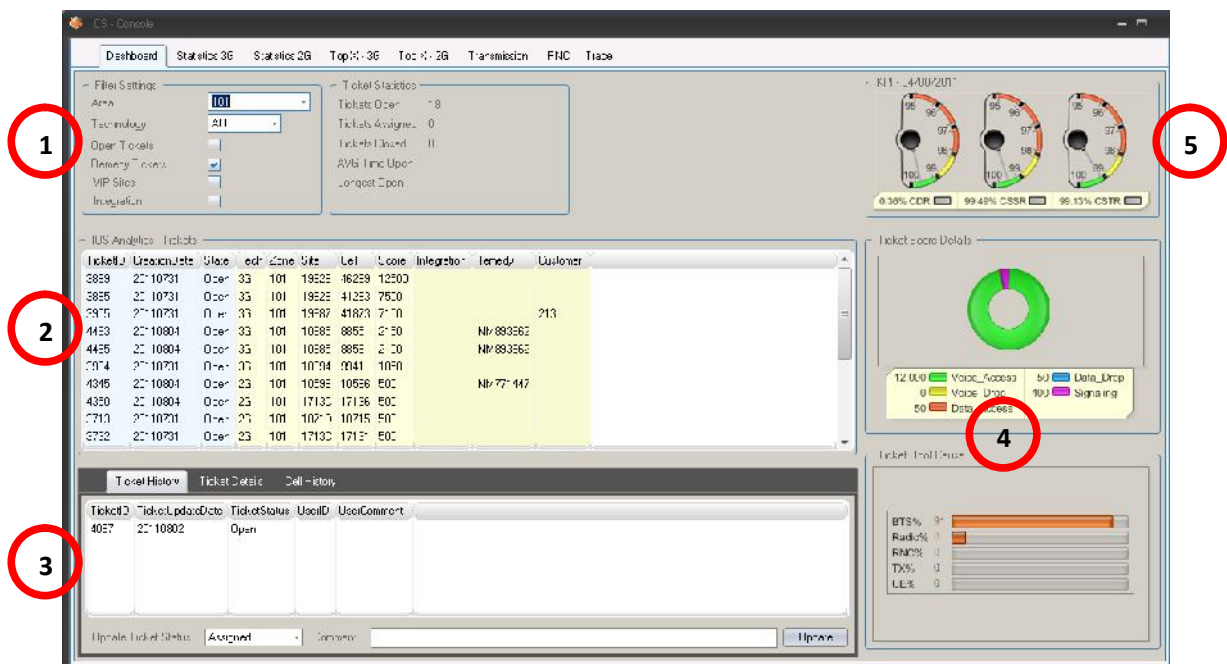


Figure 42 Dashboard Tab

1

Filter Settings are the variables that are taken into account for producing the “Ticket Statistics”, the “KPI Overview” and the list of IOS Tickets.

Field	Description
Area	Enables selection of the area of interest
Vendor	Select the vendor
Technology	Selects the technology
Open Tickets	If checked, only open tickets will be shown
Priority	Selects the priority to filter on
VIP Sites	If checked, ticket lists will be updated to show only open tickets on VIP sites

Table 4 Description of the Dashboard filter settings

Changes in the Field Settings are immediately reflected.

2

Each IOS Ticket can be read from the listbox as shown in figure below and consists of the following fields:

Field	Description
TicketID	A unique identifier for the IOS Ticket
CreationDate	The creation date of the IOS Ticket
State	The current status of the ticket (Open, Assigned, Closed)
Tech	The technology of the cell for which the ticket is generated
Vendor	
Zone	The zone of the cell for which the ticket is generated
Site	The site of the problem cell
Score	The total ticket score assigned, based on the rules breached
Priority	Indicates the priority of the ticket
VIP_Site	Displays if this site is registered as a VIP site
Customer	Indicates if the site has a VIP customer

Table 5 Field Description of the IOS Tickets

TicketID	CreationDate	State	Tech	Vendor	Zone	Site	Score	Priority	VIP_Site	Customer
121	20130730	Open	2G	NOR	0		700	1	0	0
22	20130721	Open	2G	NOR	0		450	1	0	0
38	20130722	Open	2G	NOR	0		450	1	0	0
360	20130826	Open	3G	HUA	0		350	1	0	0
9	20130721	Open	2G	NOR	0		300	1	0	0
347	20130824	Open	2G	NOR	0		300	1	0	0

Figure 43 IOS Analytics - Tickets List Box

A left click on a Ticket updates the “Ticket Score Details “ donut chart and the “Root Cause” bar chart. The ticket progress details, available under the tabs “Ticket History”, “Ticket Details” and “Cell History”, are updated.

A right click enables quick mapping of the cell, enables flagging the cell in the Statistics object tree related to the technology, enables immediate launch of the timebased statistics or, if available, visualizes the Remedy ticket information.

3

To track the IOS tickets, a straight forward mechanism has been set in the IOS Ticket History tab. A listbox shows the the actions taken on the selected ticket. By choosing the a new Status and adding a comment, the status of the ticket can be updated.

As described earlier, a ticket is created ones rules are breached. To know the rules that contributed to the score, open up the Ticket Details tab. The contents of the treelist shows only the rules that add to the score.

The Cell History tab graphs the scores per category for all tickets made in the past for the problem cell in question. The goal is to identify if the cell has repetitive problems in a certain area.

4

If a ticket is clicked, the breakdown of the ticket score is visualized using the donut chart. Five categories have been defined:

- Voice access: all scores that relate to voice access problems in 2G and 3G
- Voice drop: all scores that relate to voice retainability problems in 2G and 3G
- Data access: all scores that relate to data access problems in 2G and 3G
- Data drop: all scores that relate to data retainability problems in 2G and 3G
- Signalling: all scores that relate to signaling issues

For 3G the “Root Cause” bar chart is updated with an indication of what part of the system contributed most to the score.

5

On a daily basis the IOS Server collects aggregated KPI information for three KPI’s: CDR, CSSR and CSTR. The gauges reflect the performance of yesterday for the technology and area chosen in the Filter Settings.

2.3.2 Timebased Statistics for a Technology

The main performance monitoring functionality of IOS for e.g. 3G is located under the Stats 3G Tab where the performance of any measured object in any time resolution can be charted based on the PM Datawarehouse.

The charts come in preconfigured categories, one custom tab and a special “*CLUSTER*” tab, covering over 300 KPI’s. The preconfigured charts are defined on the IOS Server by the administrator and can be altered using the Chart Manager. All users connected to the IOS Server have the same charts.

The tab consists of 4 main parts:

- 1 - The explorer bar where objects can be searched and selected, where the timeframe of statistics can be set and where the set of Charts and KPIs to be queried can be set.
- 2 - The Source Type control, allowing to change the object on which the statistics are based on.
- 3 - The chart panels
- 4 - The info panels containing the datagrid of the chart(s), the notes made by the user, Ticket and alarm information and CM changes in the objects of interest

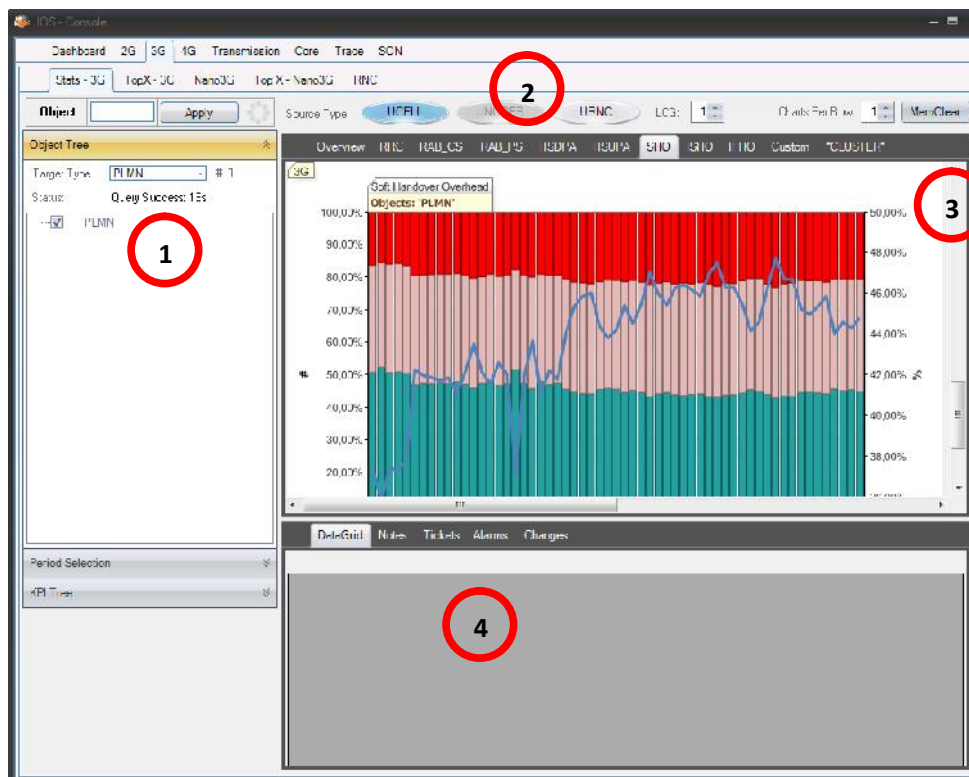


Figure 44 TimeBased Statistics - Panel

2.3.2.1 Explorer Bar

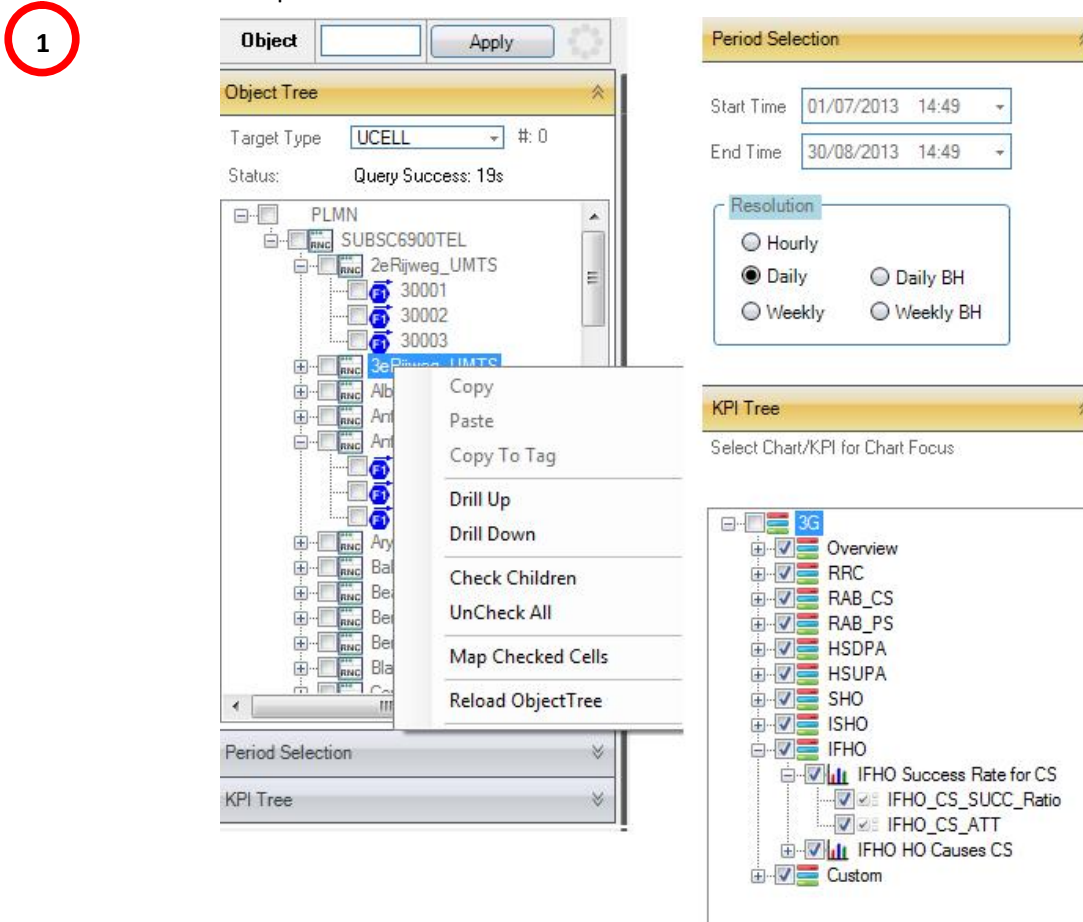


Figure 45 Timebased Statistics - Explorer Bar

Object Tree

Expanding the Object Tree of the Explorer Bar reveals the object hierarchy that was downloaded from CM data warehouse during start up of IOS.

The level shown in the object tree can be controlled using the Target Type combo box. For e.g. 3G the following types are available: WCEL, WBTS, RNC, PLMN, TAGS. Choosing WCEL will show the tree up to WCEL level. If the Target Type is changed, the object tree will be redrawn and all previous selections will be lost. The Target Type sets the object selection type for which statistics have to be generated.

Searching in the tree can be done by typing in the text box next to the Apply button. An instant search is done per character typed, matching as closely as possible the entry.

Once the object of interest is found, it must be checked to ensure it will be part of the statistics. Checking an object will automatically check its parent objects and children. Multiple selection are allowed and enable clustered object statistics.

The symbols next to the checkbox indicate the type of object: RNC, WBTS or Cell. The color of the Cell symbol reflects the band used and matches the colors used in the Map Window.

The right click context menu has the following functions:

- Copy: if objects of the target type are selected, the Copy button will be enabled and show the number of objects checked. A click will place the checked object names in the clipboard.
- Paste: objects can be pasted from for example Excel to quickly select a cluster. The procedure is to first set the Target Type, then copy from your source (usually Excel) and then right click. The context menu will show how many matches were found and if clicked, will select the matching objects.
- Copy To Tag: copy the checked objects to a Tag
- Drill Up: this function will change the objecttree target type to the object 1 level higher (parent) than the current level, and set the selection to the object 1 level higher. E.g. a selected WCEL object, drilled up, will result in a change to WBTS target type and a selection on the parent object of the selected wcel.
- Drill Down: this function will change the objecttree target type to the object 1 level deeper (child) than its current level, and select all children of the current object. E.g. a WBTS selection , drilled down, will result in target type set to WCEL and all children selected of this WBTS selection.
- Check Children: this function will check all the childs of the highlighted parent
- Uncheck All: this function removes all the checks
- Map Checked Cells: Maps the checked cells by searching and selecting each cell in the Map Window. If cell isn't found, a warning message will be shown. This only works for CELL type objects.
- Reload Object Tree: This function re-queries the CM data warehouse to get the latest version of the object tree for this particular technology. By default, IOS loads all object trees on startup, however, to refresh without startup, this function is recommended.

Period Selection

Expanding the Period Selection explorer bar item reveals the settings for the time based statistics. A start date/time and end date/time can be set together with the resolution.

In the Resolution groupbox, the interval of the smallest datapoint can be set, but also selects the source of the database.

Currently 5 datapoints have been defined: Hourly/Daily/Busy Hour/Weekly/Weekly Busy Hour

Remark:

- When selecting Hourly, the start and end date are set to Yesterday and Now, covering 1 day of hourly data
- When selecting Daily, the start and end date are set to Now-60 days and Now, covering 60 days of daily data

KPI Tree

The KPI Tree is the hierarchy structure of the tab and chart configuration. It displays what charts are under each tab but also what KPI's are bound to each chart.

- Clicking on a chart or KPI will jump directly to the right chart.
- Checking or un-checking a chart will either include or exclude a chart from the query that is fired when the Apply button is pressed.

Tip:

In environments with many cells and large timeframes, it is recommended to make a subselection of the categories or charts. This will speed up response times dramatically.

2.3.2.2 Source Control

The source control, which is usually left untouched as it is set automatically if a certain target type is chosen, defines what basis object is used to create the statistics. E.g. If WCEL is toggled, the statistics will be based on measurements collected by the WCEL object.

The Local Control Group (LCG) is a special selector in case there are more than 6 cells in a WBTS. As some vendor system splits the WCEL based statistics in 2 LCG's, a selection needs to be made to get the statistics of the intended cell(s). The default value is 1.

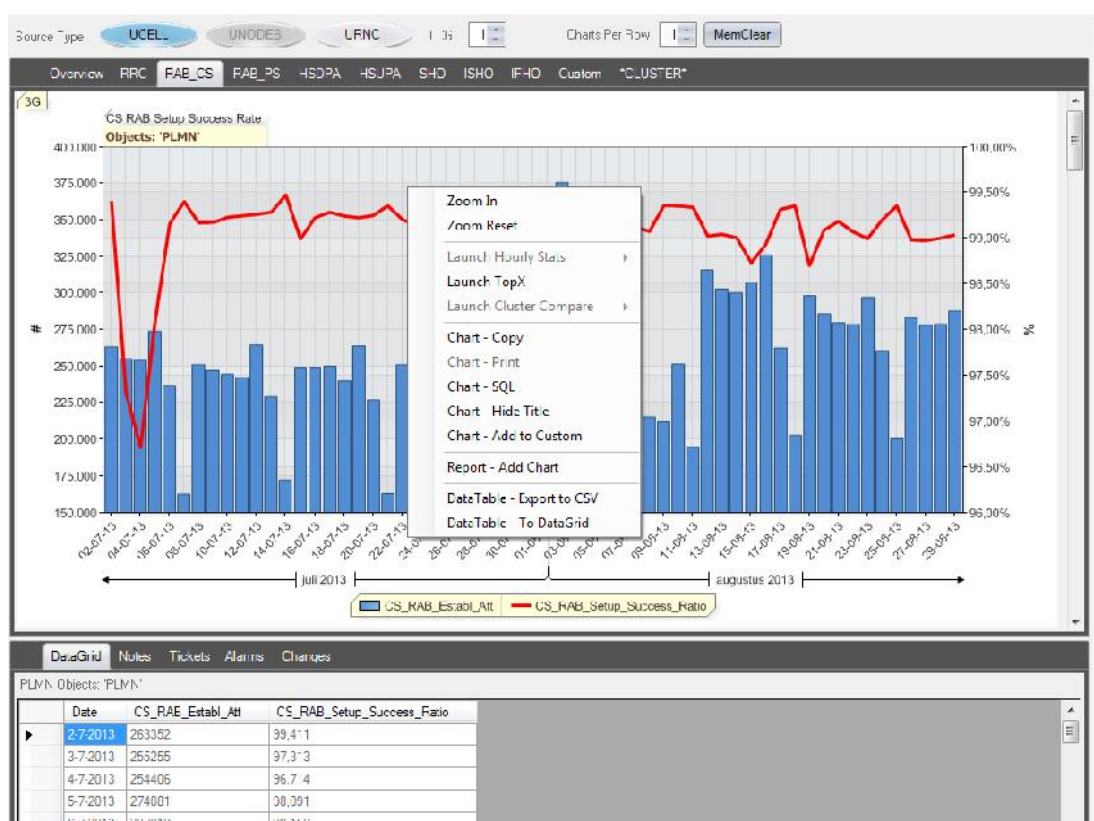


Figure 46 TimeBased Statistics - Chart Tabs

Charts per Row sets the number of charts that are visible in on the screen. Default value is 1. On larger screen, increasing this number utilizes the screen estate better, by rearranging the charts from left to right, with number Charts per Row number representing the number of charts that are displayed per visible row.

The MemRelease button drops and redraws the charts, releasing chart cache information from the memory. Having substantial number of charts open on a machine that has limited RAM resources, can slow down the overall performance.

2.3.2.3 Tab Control – Charts

The performance charts are categorized in tabs, as configured on the IOS Server, except for 2 tabs: Custom and *CLUSTER*.

The organization of charts in tabs and the charts itself can be done in the Chart Manager as described in 2.2.6.

The Custom tab holds all the charts that are custom made by the user, entered through the “Chart Manager” - form. Only the charts made by the user itself will be visible. A user cannot see Custom Charts made by another user.

The *CLUSTER* tab holds all the charts that are the result from the special “Launch Cluster Compare” function in the right click menu of the chart.

The right click menu of a chart has the following functions:

- Zoom In: A toggle button that activates the zooming function for the charts. If active, a left click and drag will draw a rectangle which sets the X and Y axis boundaries on which the chart will zoom in.
- Zoom reset: brings the chart back into original state
- Launch Hourly Stats: obsolete
- Launch TopX: When multiple objects are charted, the user can launch a TopX list for these objects on a day of interest. By right clicking on the chart element, the TopX will be launched for the timeunit that was on the x-axis of the chart element that was right clicked.
- Launch Cluster Comparison: When multiple objects are charted, the user can make a comparison of these multiple objects for a single KPI in time. A new chart will be drawn in the *CLUSTER* tab of the technology of interest.

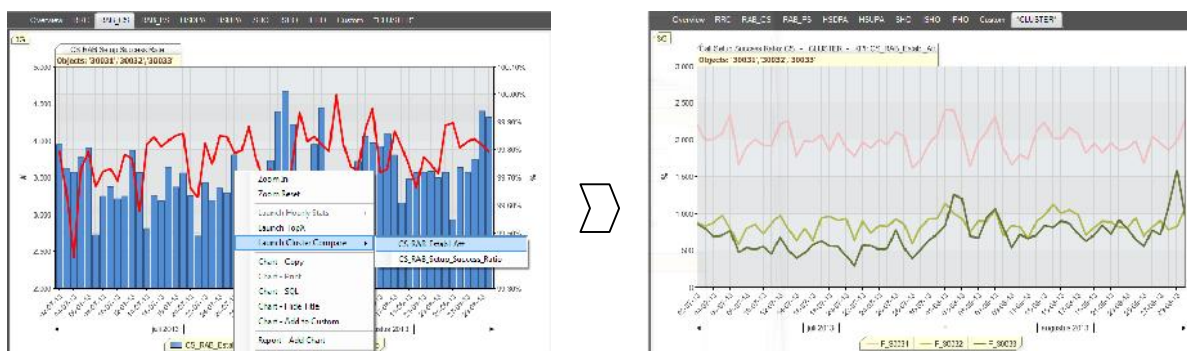


Figure 47 TimeBased Statistics - Cluster Comparison

Right clicking on a chart element and selecting Launch Cluster Compare, will provide a dropdown box of all KPIs that are present on that chart. Selecting a KPI, will then query and draw the result of that KPI for the objects selected in the tree and for the period set in the period selection. An “F_” prefix is set in front of the object name.

Additionally, a visual and geographic comparison can be generated by clicking on the cluster comparison chart. A black axismarker will be drawn on the chart, but also the cell objects that are part of the query will be selected on the map and a thematic is applied with default color Grey (low value) to Red (high value).

Pressing PgUp or PgDown with the chart in focus, result in the next time element to be selected or the previous time element.

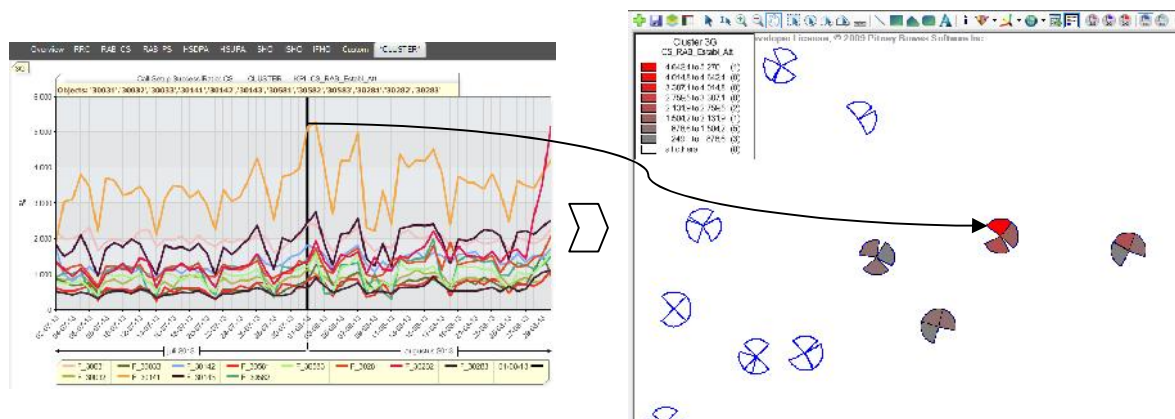


Figure 48 Timebased Statistics - Cluster Comparison and Mapping

- Chart – Copy: makes a bitmap copy of the chart and places in the Clipboard
- Chart – Print: currently not available
- Chart – SQL: dumps the SQL for that chart in the Clipboard
- Chart-Hide Title: If many objects are selected, the chart title can affect the esthetics of the chart itself. Hiding the title of the chart solves that problem.
- Chart – Add To Custom: This function enables to add a chart to the personal Custom Tab, taking over the entire chart configuration. For further adaptation to the chart, go to Chart Manager.
- Report – Add Chart: This function enables to add a chart to a Report as defined in chapter 2.2.1. All reports that the user has under his control will be shown. Selecting a report will add the right clicked chart to the bottom of the existing report.

- DataTable – Export To CSV: Opens up a dialog box in which a user can enter a filename to save all KPI results for that particular technology and object selection into CSV format.
- DataTable – To DataGrid: Fills the datagrid with all KPIs of the selected technology and objects.

A left mouse click on a chart will update the datagrid below the chart with the raw data of the chart, enabling the user to copy/paste this data.

2.3.2.4 Tab Control – Info

Data grid

DataGrid Notes Tickets Alarms Changes				
UCCELL Objects: '30031','30032','30033','30141','30142','30143','30581','30582','30583','30281','30282','30283'				Observed Objects
	DATE	RRC_Att_Serv	RRC_Setup_Success_Ratio_Service	KPI Names
▶	2-7-2013	169126	99,822	
	3-7-2013	157260	99,889	
	4-7-2013	166678	99,809	
	5-7-2013	167287	99,894	
	6-7-2013	159985	99,665	

Figure 49 TimeBased Statistics - DataGrid

The DataGrid has two states:

- It shows the data series of chart that has been clicked
- It shows the entire KPI dataset for all charts of that technology.

Sorting can be achieved through clicking on the headers. Copying to Clipboard is done by selecting the records of interests and pressing Ctrl-C.

Notes

A user can store notes about a particular object (a single cell, or a group of cells or rnc,...) on the IOS Server. This is means to track changes or important events in the network and share it amongst the user base. Examples of useful notes to the engineer are: a RNC software upgrade, a feature trial, an optimization campaign in a certain region, etc..

A note is represented in 2 ways: it is shown in the datagrid that is located in the Notes tab, and, it is drawn as an x-axis marker on each chart.

Each note contains the following properties:

- A timestamp, indicating the time and date to which the note relates
- A user, indicating whom created the note

- A technology, indicating to what technology the note relates
- A Note Type, is a category that relates to this note. Categories are defined by the system administrator. Each category has its own color representation in the grid and on the chart.
- Department: is the department that issued the note. Departments can be defined by the system administrator.
- Short and long description
- A list of objects that is bound to this note.

All users can view the notes.



Figure 50 Timebased Statistics - Notes Description

To add a new note, press “Add Note” and a new dialog box will appear. By default the Notes dialog box will read the current object selection based on what target object type is selected and what objects are selected in the object tree. This is summarized in the Current Object Selection Groupbox.

The user can set all desired parameters of the note and hit Submit to commit the note to the server.

The screenshot shows a 'Notes' dialog box with the following sections:

- Current Object Selection:**
 - Technology: 3G
 - Object Type: UCELL
 - Object selected: 4
- Objects Affected:**

Object	Type
30031	UCELL
30032	UCELL
30033	UCELL
30061	UCELL
- Note Details:**
 - Date of Change: 02/09/2013 10:19
 - Username: Guy
 - Type of Charge: Feature Pilot
 - Department: Optimization
 - Short Description: (empty text box)
 - Long Description: (empty text area)

A 'Submit' button is located at the bottom right of the dialog.

Figure 51 Notes - Additions

Get Notes retrieves the Notes from the IOS Server. The following notes will be fetched:

- Notes that have same object type and same object name as the current selection in the object tree
- Notes of parents of the selected objects in the object tree. E.g. if an RNC has a note, then this note will always appear on cell based selection of this RNC.

Modifying a note can be done through double clicking on the note. The dialog box will appear and changes can be applied

Deleting a note is achieved by selecting the entire row in the datagrid and pressing "Del" button.

Tickets

If the IOS solution is interfacing with a third party ticket system, then this function will be enabled.

For the objects selected in the Object Tree, the history of all Third Party tickets can be fetched through "Get Tickets". The high level Third Party ticket information is then displayed in the grid. A double click on the ticket number launches the Ticket Window with the detailed ticket information (if available). (See figure 29)

The dates of each ticket and its severity is also displayed on the chart as a vertical line. Severe problems are represented in red whereas low prio problems have a light orange color.

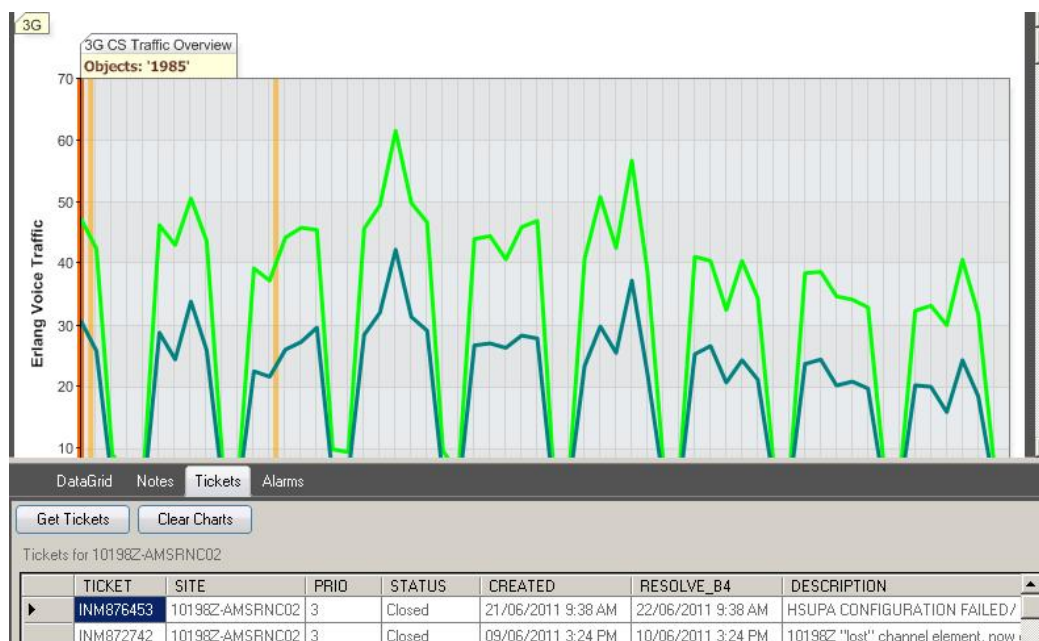


Figure 52 TimeBased Statistics - Ticket Grid

Alarms

If the IOS solution is interfacing with an Alarm database or is configured to collect Fault Management information, then this feature will be enabled.

Alarms stored in the live OSS or datawarehouse can be seen in the grid, just like the tickets in previous paragraph. The alarms are also plotted on the charts as vertical lines, clearing first other vertical indicators.

Changes

The Changes tab allows for quick correlation of PM data with CM changes in the selected objects. CM changes can be tracked on target object types and parent object types. Any parameter change that happened, will be displayed in the grid, and displayed as a black vertical marker on the chart containing a label: "parameter: FROM x to y. "

Tip:

It is recommended to query for CM changes 1 object selection at a time, to avoid information overflow. This can easily be achieved by selecting/deselecting the object of interest in the object tree, without requerying for PM data.

To see the changes of 1 particular parameter, double click on the parametername in the grid.

To see the entire parameter set of the object type that has been requested, press the UnHide button.

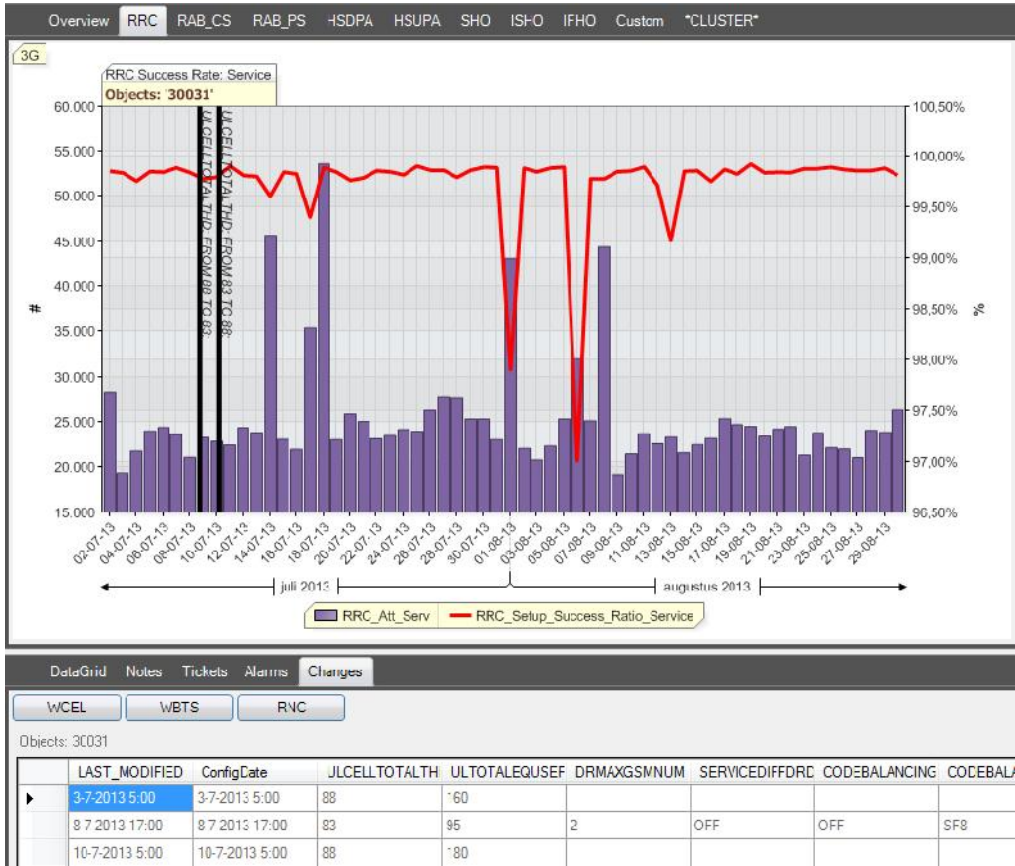


Figure 53 TimeBased Statistics - Changes Example

2.3.2.5 Generic Workflow

Workflow – Generate WCEL statistics:

- Set target type to WCEL

- Select objects:
 - o Manually
 - o Through Pasting a set stored in the clipboard
 - o Through sending from the map
 - o Through sending from TopX
- Set time interval and resolution
 - o Default value is 60 days and daily
- Press Apply
 - o The map will synchronize with the objects selected if object is a WCEL or WBTS.
 - o A multi-threaded query will fire to get the data from the datasource, disabling the controls of the 3G tab. The other tabs are however available for other actions.
 - o The process is complete if circular progress bar has stopped.
 - o At any time, the “Abort” button can be pressed to stop the query.

2.3.3 TopX Statistics for a Technology

Ranking objects based on KPI's in a certain object set can quickly point towards an isolation of a bad performing object. The TopX module, available in 2G, 3G and 4G, enables to execute such a TopX request in a straight forward manner, based on the same KPIs as in the timebased technology modules.

The tab consists of 3 main areas, the Object Tree, the Source Type selection and the Charting panels.

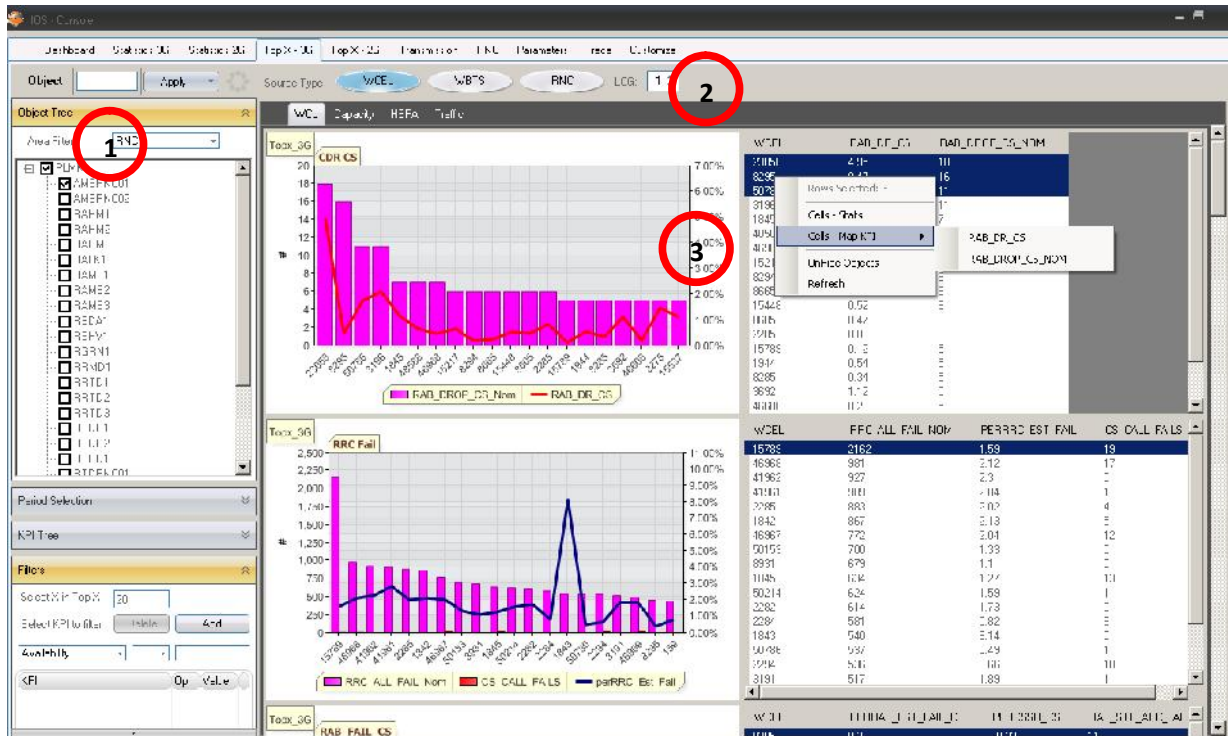


Figure 54 TopX - 3G Panel

1

The same functionality as in the Object Tree for Statistics 3G is present. Refer to chapter 2.3.2. The Period Selection is identical to the Statistics 3G Period Selection. However, the default value is set to 1 day, yesterday.

In the Filters explorer bar item, the number X of TopX can be set. Default value is 20. Additionally it is possible to add filters based on KPIs to limit the outcome of the TopX. As an example: to filter on Availability < 99%: set KPI to Availability, choose the "<" sign and set the value in the last text box and click add. Running the statistics will now filter the results in all charts where the Availability KPI is used.

2

The source control, which is usually left untouched as it is set automatically if a certain target type is chosen, defines what basis object is used to create the statistics. If WCEL is toggled, the statistics will be based on WCEL object statistics.

The Local Control Group (LCG) is a special selector in case there are more than 6 cells in a WBTS. As the NSN system splits the WCEL based statistics in 2 LCG's, a selection needs to be made to get the statistics of the intended cell(s). The default value is 1.

3

As in Statistics 3G, the charts and accompanying tables are categorized. The configuration comes from the IOS Server and is defined by the Administrator, but can be changes in the Chart Manager. There's also a Custom Tab, that holds the user generated charts.

Each chart has a datagrid shown next to it, containing the same information as the chart. The charts have the same right click context menu as described in chapter 2.3.2.3, except for:

- Object DrillDown: When the user right clicks on the chartelement of a certain object, the user can decide to fetch underlying statistics of the children of the object. E.g. when a drilldown on a WBTS object is performed, the statistics will be requested for WCEL for that particular object.

The grids also have a context menu as shown in Figure 30:

- Cells Stats: Sends the highlighted objects to the Statistics 3G panel and runs the statistics. A fast method to go from TopX mode to time based statistics mode.
- Cells – Map KPI: pops up an additional menu showing the KPIs involved in the chart. Clicking on the KPI of interest will map the cells that are highlighted and a thematic layer will be added showing a gradual symbol in red for every highlighted object. The size of the red gradual symbol gives an indication about the size of the problem versus the other cells. A large red dot hence means a bigger problem.

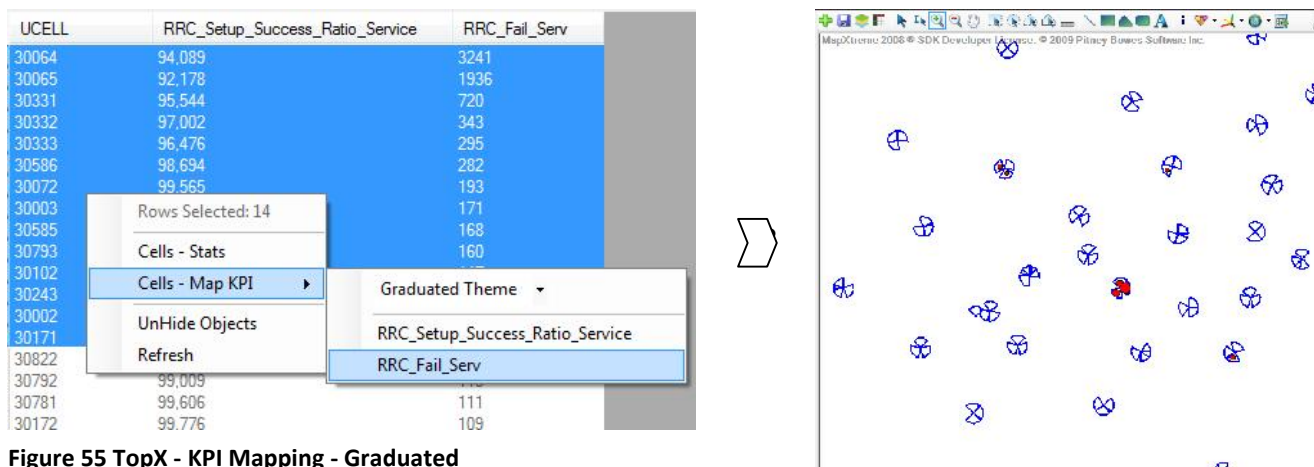


Figure 55 TopX - KPI Mapping - Graduated

The benefit of a graduated symbol over a ranged theme is that the graduated symbol automatically sized if the user zooms in or out. When the map is zoomed out a lot, it is easy to spot the object with the largest dot.

When the user is focusing on a particular cluster, the ranged thematic might provide a better visualization:

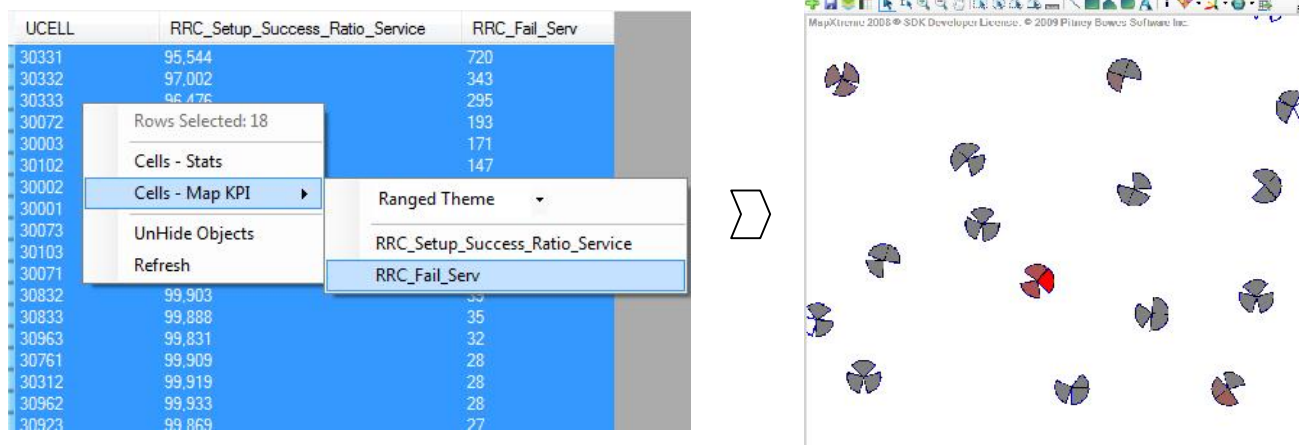


Figure 56 TopX - KPI Mapping - Ranged

The Mapinfo Layer is named: "TopX_<technology>_<kpi_name>_Map"

- UnHide Objects: expands the datagrid to show parent object names like RNC and WBTS description data
- Refresh: Not used

2.3.4 Trace Module

Prerequisites:

Trace License is available

NSN Networks interfacing with NSN OSS.

The trace module is an NSN specific Trace Visualization module. The module collects a selection of trace information coming from the NSN NetAct TraceViewer. The trace data is must be collected within NSN NetAct and the IOS client must have access to the NSN NetAct OSS via ODBC.

The trace module consists of 6 report types, represented as grids, a tab containing timebased charts of key radio elements and a functionality to synchronize with the map.

Workflow:

- Activate a trace in NSN NetAct, ensuring at least RAN_IntraFreqMeas report is measured
- Once trace is finished, press the Refresh button to get the latest TraceIDs that are currently in the NSN NetAct system
- 1** - Select the TraceID in the combobox and press Get Trace
 - 2** o 6 Grids will be filled with radio specific reports
 - 3** o 4 Charts are available in the Chart Tab:
 - RSCP measurements by the traced mobile in time
 - The best RSCP measurement in time, with the associated CellID in the axis
 - EcNo measurements by the traced mobile in time
 - The best EcNO measurement in time, with the associated CellID in the axis

Remark: Charts and Map function is only available if RAN_IntraFreqMeas report has results.

- Press Map to visualize all the Cells that have been detected in the active set of the traced UE. This is based on the CELL_1, CELL_2, CELL_3 values in the RAN_IntraFreqMeas.

TRACE REFERENCE	Date	SD1	SD2	SD3	ECNO1	ECNO2	ECNO3	RSCP1	RSCP2	RSCP3	CELL 1	CELL 2	CELL 3	Pwr3GLevel	Pwr3GQuality
1	2-8-2013 12:35:47	110	97		-3.5	-7		-87	-83		10156	10154		-106	-15
2	2-8-2013 12:35:49	97	110		-9.5	-11.0		-90	-91		10154	10156		-106	-15
3	2-8-2013 12:35:49	110			9						10156			106	15
4	2-8-2013 12:35:49	110	97		-9.5	-9.5		-95	-94		10156	10154		-106	-15
5	2-8-2013 12:35:49	110			-9.5			-95			10156			-106	-15
6	2-8-2013 12:35:49	110	97		-7.5	-11.5					10156	10154		-106	-15
7	2-8-2013 12:35:49	110	97		-6	-15.0		-87	-97		10156	10154		-106	-15
8	2-8-2013 12:35:50	110	97		6	15.5		87	97		10156	10154		106	15
9	2-8-2013 12:35:51	110	97		-6.5	-12		-91	-102		10156	10154		-106	-15
10	2-8-2013 12:39:20	110			-3						4513			-106	-15
11	2-8-2013 12:41:51	110			-3.5						4513			-106	-15
12	2-8-2013 12:39:37	110			-4						10153			-106	-15
13	2-8-2013 12:39:38	110			3.5						10153			106	15
14	2-8-2013 12:40:21	110	97		-5.5	-8.5		-84	-88		10156	10154		-106	-15
15	2-8-2013 12:43:06	110			-3						4513			-106	-15
16	2-8-2013 12:43:06	110			-7.5			-10			4513			-106	-15
17	2-8-2013 12:43:06	110			-2.5			-76			4513			-106	-15
18	2-8-2013 12:43:09	110			6						10156			106	15
19	2-8-2013 12:43:09	110			-5			-79			10156			-106	-15
20	2-8-2013 12:43:09	110			-5			-79			10156			-106	-15

Figure 57 Trace Module - Overview

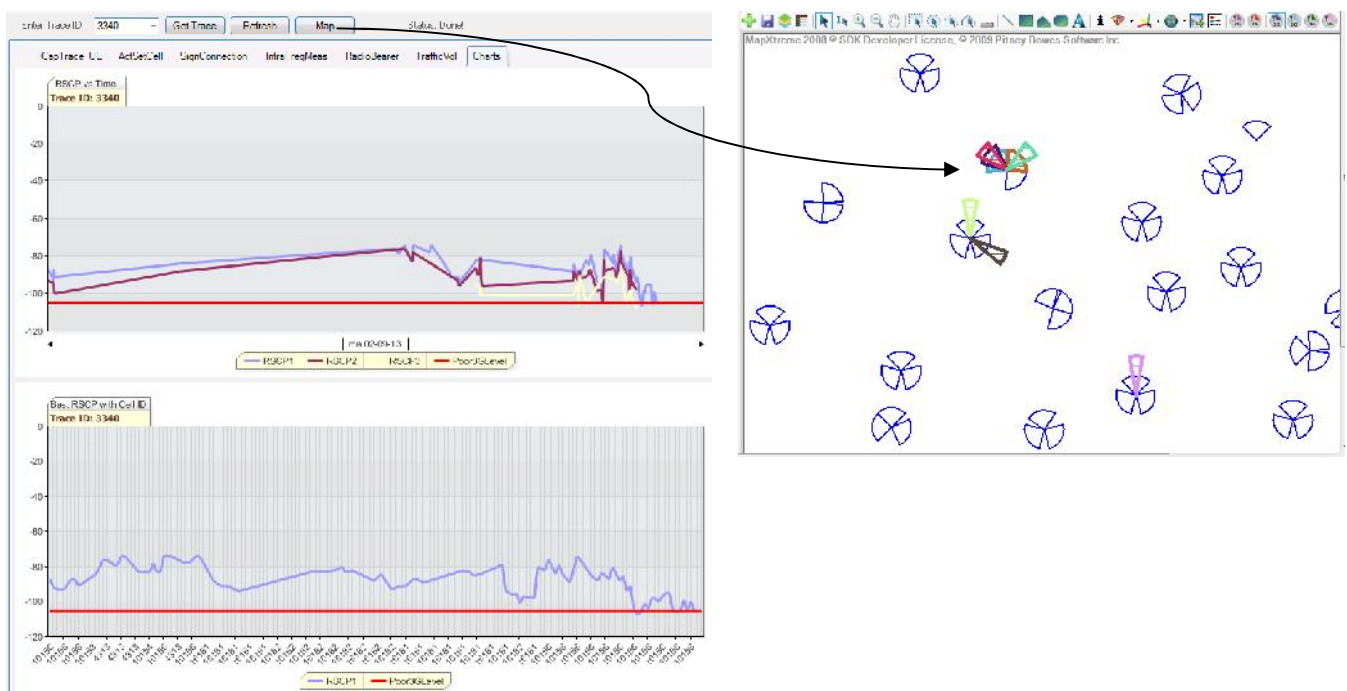


Figure 58 Trace Module – Charts and Mapping

2.3.5 SON Module

Manual of SON is on request.

3. Work Flows

3.1 Purpose

The purpose of this document is to illustrate functions of the tool in context of several example workflows and analysis tasks which may be encountered in a normal day to day working environment as an optimization engineer.

The workflows are meant to consolidate the understanding of how the tool can be used in different ways to troubleshoot and investigate Network issues, customer complaints and other analysis tasks

3.2 Cluster Performance Check

Objective:

To verify the proper performance of a cell cluster. This could be performed after a network change or it can also be used with newly integrated sites (e.g. production line)

Outcome:

The tool helps to check that KPI's are within target and identifies poorly performing cells in lists, charts and geographically on map

Step by step:

1. If the source is a list of cells or sites, paste them directly into the object tree of the console window
2. Run combined daily statistics for the last 2 weeks
3. Review KPI verify that performance target is met for entire cluster as a whole
4. By now the sites will be highlighted on map. Check on map that location is as expected and indeed a complete cluster has been supplied.
5. Check if any alarms are present on any of the sites in the cluster. If there are alarms , ensure they have been acknowledged. If an alarm is present the "DN" field shows which cell object is affected
6. Run all highlighted Cells of the Cluster through the TopX generator: right click map, send cells to TopX. Chose several days as the time period. Chose KPI filter thresholds if needed, e.g. CSSR < 98%
7. Check first Availability, CSSR, CDR then move onto capacity and HSDPA page.
8. Identify any outliers and check their individual performance, create ticket or take action as necessary
9. Once cluster has been reviewed and sites with problems have been actioned, the report generator could be used to Create a final Cluster performance report

3.3 Customer Complaint Analysis – General Voice

Objective:

To investigate a customer complaint from both Network performance and customer call records point of view.

Outcome:

Identify if the root cause of the complaint is Network performance or coverage related and help the optimizer decide on what actions are needed to solve the customer's issues

Step by step:

1. Use map search to view the postcode where the problem occurs
2. Check coverage layer or Drive test layer, open relevant tab files and check Cell footprint tool - coverage issue plausible?
3. Map CS calls by using IMSI in the "Customer" field, left pane of map view. Check map for unsuccessful calls.
4. Show grid at using the button at the top of map view. Check clear codes in the grid. Also browse grid to check if customer is using mainly 2G or 3G
5. Check on map for missing neighbors, by right clicking sectors and making relevant selection. Check outgoing and incoming.
6. Check parameters of sites are normal, e.g. check power settings, cell bar settings, and minimum access settings. Check also tilt settings
7. Select the cells on map that are mostly used by customer – right click and send to Console to analyze combined performance. Is general performance in the area good?
8. Send cells to TopX – identify if there is a particularly poor performing cell or a cell with outage of fault during time of complaint
9. Check tickets and alarms.
10. Using all the info gathered carry out necessary action to solve issue. If no fault is found from IOS study then a Drive Test might be necessary

3.4 Customer Complaint Analysis – PS, Low throughput

Objective:

To more specifically identify why a customer may experience low throughput.

Outcome:

Help identify the root cause of the low throughput problem

Step by step:

1. Use map search to view the postcode
2. Check coverage layer or Drive test layer - coverage issue plausible?
3. Map PS calls. Find the cells the customer uses mostly
4. Check in the PS call record volume indicator if the customer ever achieves high download volumes. Look for the highest volume cell (largest blue circle) and use the layer information button to see what data quantities were transferred and on what technology (2G/3G)
5. If customer is always on 2G, but has a 3G handset (check with TAC) , handset may be locked to 2G. Retrieve calls from some time ago and check in the filter section of the map grid if the customer never accesses 3G
6. Check if 3G cells in the area are operational and performing normally. Send cells to TopX and check HSDPA usage and success statistics
7. Check if 2G to 3G neighbours are defined, so reselection to 3G can occur
8. If customer toggles between 2G and 3G a lot he may be in a marginally covered 3G area. Conduct a subscriber trace if possible to see RSCP, EcNo, Scrambling Code, propagation delay and if the handset is using HSDPA or not.
9. Check tickets and alarms.
10. Using all the info gathered carry out necessary action to solve issue. If no fault is found from IOS study then a Drive Test might be necessary

3.5 Performance check: top to bottom drill down

Objective:

Occasionally the PLMN, RNC stats experience degradation. This drill down procedure will quickly identify which sites are causing this and what the root cause is according to NSN counters

Outcome:

Quickly identify the nature and severity of a problem affecting PLMN. Create analysis which sites are affected and what is the root cause.

Step by step:

1. Show long term PLMN or RNC stats. Verify that there is indeed degradation. At this point already note down the failure cause that seems to be causing the degradation
2. Go to TopX , select PLMN in the object tree and select source button “RNC”. Chose a day where the degradation was significant
3. RNC’s will now be ranked according to highest failures. Check the KPI “box” that has the highest failures according to step 1. It will now become apparent if the issue comes from just one RNC, several or all of them
4. Still in TopX, select the worst RNC’s from the object tree and as the source select WBTS. Chose Top 50 on the left hand pane.
5. Check the resulting ranking chart for the KPI in question to see if the failures are coming from a small number of sites.
6. To geographically visualize these sites by selecting them in the TopX, rightclick and show on map, according to the desired KPI. KPI can be first sorted if needed.
7. Send the worst few sites to the console and view time based charts.
8. Check alarms, tickets and use hourly stats to check if the problem is still ongoing.
9. Using all the info gathered the following is now possible to create a management report using the following details:
 - a. Where geographically the problem is
 - b. How long it’s been going on for
 - c. How many RNC’s are having high contribution
 - d. The cells and site ID’s that are failing particularly high
 - e. What the NSN failure cause counter is
 - f. Are these cells being dealt with (having alarms or tickets) or do actions need to be taken

3.6 Correlate performance issue with Parameter settings

Objective:

Since the tool is able to produce cell statistics as well as map parameter values it is sometimes interesting to correlate these

Outcome:

Find if a correlation exist between a certain KPI and a parameter setting

Step by step:

1. Create TopX for Cells from an RNC
2. Plot cells on map with some performance issue, eg HSUPA max user blocking
3. On map, use info button to and select any cell to show parameter values
4. Map an interesting parameter that has relevance to the issue, eg HSUPA X-users
5. Once the parameter is mapped it is easy to visually correlate if cells with HSUPA max user blocking also have a low setting for HSUPA X-user parameter

3.7 Find Cells with overshoot issues

Objective:

Use the missing neighbor and traffic distance feature to check if a cell is overshooting

Outcome:

Being able to identify an overshooting cell

Step by step:

1. Create TopX for Cells with high uplink interference or chose a high site
2. Locate the cell on map
3. Right click the cell and show traffic distance based on propagation delay
4. Find 3G to 3G missing neighbours
5. If uplink noise is high, propagation delay reaches far and there are high handover attempts to missing neighbours far away, then the Cell is likely to be overshooting and needs a downtilt

3.8 Select cells with specific parameter criteria in a specific area

Objective:

Select a group of cells of interest using

Outcome:

Example: find EGSM cells with Cell bar active

Step by step:

Given a parameter or Frequency criteria, select cells from map

1. Map parameter value of interest e.g. Cell Bar
2. Show only Frequency layer if interest, e.g. EGSM
3. Select cells from map matching criteria using polygon tool, these can be used externally or sent to stats or TopX console

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